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All the topics of this UPSC IAS Edition are directly or indirectly important for the prelims & main examination.
There are some topics which can be coded in answer writing of other topics in the main exam.

BRENDAN LYNCH LEADS 4-MEMBER TEAM

US team in Delhi today, urgency to finalise deal

Trade talks come as new tariffs set to replace reciprocal tariffs in July

Ravi Dutta Mishra
New Delhi, May 31

A TEAM of United States Trade Representative (USTR) officials, led by chief negotiator Brendan Lynch, is set to commence negotiations with India over four days starting Monday to finalise an interim agreement.

Government sources said there is a sense of urgency now because a new US tariffs architecture is likely to replace the reciprocal tariffs announced by US President Donald Trump, and deemed illegal by the Supreme Court in February. The new rates under Section 301

»CONTINUED ON PAGE 2

E. EXPLAINED

Key to investments

For India, finalisation of an interim trade deal is seen as important in restoring confidence among investors. Also, there is a sense of urgency since a new US tariffs architecture is likely to replace reciprocal tariffs before July-end.

THE IDEAS PAGE P 11

INDIA-OMAN PACT HOLDS PROMISES FOR FARMS, FACTORIES & SMALL BUSINESSES

NATION

NEPAL HAS ALSO ENCROACHED ON INDIA'S TERRITORIES: PM BALENDRA

PAGE 9

LLOYDS METALS PROJECT IN GADCHIROLI

Maharashtra clears mine project, says not in tiger area; maps show otherwise

Jay Mazoomdaar
New Delhi, May 31

THE MAHARASHTRA government on May 13 exempted an iron ore mining and processing project in Gadchiroli from wildlife clearance by claiming — incorrectly — it was not located in any tiger corridor, official communications reviewed by *The Indian Express* show.

The project proposal involves diverting 9.4 sq km of forest land in Gadchiroli for iron ore extraction and processing by Lloyds Metals & Energy.

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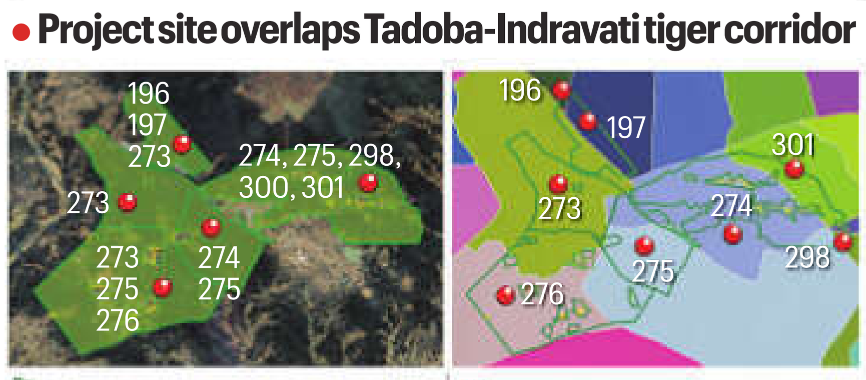
● **Project site overlaps Tadoba-Indravati tiger corridor**



Proposed project site
Forest compartments 196, 197, 273, 274, 275, 276, 298, 300 and 301 cited as part of the project

Forest compartments 196, 197, 273, 274, 275, 276, 298 and 301 marked on Tadoba-Indravati corridor in the NTCA-approved TCP

SOURCE: PARIVESH/MOEF AND NTCA

Govt seeks to bring back delimitation Bill, push joint polls by 2029

Liz Mathew & Arun Janardhanan
New Delhi, Chennai, May 31

IN THE wake of its defeat in Parliament over its attempt to push the delimitation Bill, moves are on within the Government to revive the legislation and also bring in the One Nation One Election Bill — both before the 2029 Lok Sabha elections, *The Indian Express* has learnt.

To this effect, emboldened by its victory in West Bengal, sources said, the BJP is reaching out to regional parties

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IDEA EXCHANGE

"FTAs with Europe, UK good moves... should ensure industry doesn't put pressure to take rigid positions"

MONTEK SINGH AHLUWALIA
ECONOMIST, PAGE 15

[FROM PAGE 1 | FULL REPORTS ON WWW.INDIANEXPRESS.COM]

US team in Delhi today, urgency to finalise deal

will come into effect on July 24.

Months of tension between New Delhi and Washington had resulted in steep 50% tariffs initially on India and an outflow of investments. The depreciation in the domestic currency that began due to steep tariffs last year has intensified in the backdrop of the energy shock following the war in the Gulf. New Delhi is also keen certainty on the US trade deal will restore confidence among investors.

While trade tensions have somewhat eased between the two countries, with the US reducing the reciprocal tariffs from 50% imposed in August last year to 18% in February this year, the negotiations for a deal remain challenging for India due to what officials describe as "unconventional US demands".

A senior government official told The Indian Express that the current 10% across the board tariff without a deal is a preferable scenario for India than signing a deal because of US demands that interfere with sovereign decisions on the purchase of oil and sensitive agricultural items.

The US had in April launched two Section 301 probes on India in relation to "structural excess capacity" and "forced labour". While talks have been ongoing, officials in Delhi responded to these changes, stating that surpluses are a natural consequence of global trade and a byproduct of generalised economic conditions. Further, they said India has ratified the Forced Labour Convention, 1930 and the Abolition of Forced Labour Convention, 1957, which require the prohibition of forced labour in all forms.

Near the finish line

The trade deal talks starting here could finally put an end to months of uncertainty, as USTR Jamieson Greer said last week that he expects to meet Commerce Minister Piyush Goyal "soon" to finalise the India-US framework deal. "On New Delhi, we have had many meaningful discussions. We have a framework agreement with the Indians. I have a team going there next week. I expect to meet my counterpart soon as well. I'd really like to be in a position to finalise our agreement based on the joint framework agreement we agreed to," Greer said at a Council of Foreign Relations conference.

Sergio Gor, US Ambassador to India, last week said Washington expects that a bilateral trade deal with India will be signed in the "next few weeks and months". He indicated that only "1 per cent" of the deal was left to be concluded, and negotiators were working on it. Describing US Secretary of State Marco Rubio's four-day visit to India which ended May 26 as "substantive", Gor, addressing the US-India TRUST Initiative event at IIT Delhi, said, "Just last week, India had sent a team to Washington DC to finalise the last 1 per cent of that trade deal. Next week we will welcome a US delegation here to continue those talks."

Ahead of the key talks, New Delhi is signalling greater openness in trade diplomacy. Commerce Minister Piyush Goyal last week led a delegation with 100 industry leaders to Canada, aiming to sign an agreement by the end of the year. This comes amid testy trade ties between Canada and the US during the ongoing United States-Mexico-Canada Agreement negotiations. India is also set to operationalise its trade agreement with Oman on Monday.

Uncomfortable US demands

While a government official earlier this month said India will continue purchasing Russian crude oil regardless of whether the US extends sanctions waivers, the US had imposed additional 25% tariffs on India for the purchase of Russian oil under the deal. Trade data showed that India's purchase of Russian oil began declining before recovering from March in the backdrop of the war in West Asia.

Demands relating to agricultural products have also been one of the biggest pressure points. The India-US joint statement said India will eliminate or reduce tariffs on all US industrial goods and a wide range of US food and agricultural products, including dried distillers' grains (DDGs), red sorghum for animal feed, tree nuts, fresh and processed fruit, soybean oil, wine and spirits, and additional products.

Indicating signs of disagreement, the US later revised a factsheet softening its claims about the gains it had secured from New Delhi and entirely dropping a section on digital services taxes. The earlier version of the factsheet said India had "committed to" buying more American products and purchasing "over $500 billion of U.S. energy, information and communication technology, coal, and other products". The updated factsheet, as well as the joint statement, tempered the wording from "committed" to "intends".

FDI & currency pressure

A slowdown in investments has resulted in a rapidly weakening currency. India saw gross Foreign Direct Investment (FDI) inflows rise to a new record high of $94.53 billion in 2025-26, up 17% from the previous year, although the net figure was a mere $7.65 billion, data released last week by RBI showed. The domestic currency has breached the Rs 95 per dollar level, becoming one of the worst currencies among its Asian peers.

Maharashtra project

• TIMELINE

- **DEC 12, 2023:** Applied for Forest clearance
- **MARCH 20, 2024:** Applied for Wildlife clearance
- **APRIL 28, 2024:** Applied for Environment clearance
- **MAY 12, 2024:** In-principle FC approved
- **SEPT 20, 2025:** EC approved
- **OCT 6, 2025:** Applied for amended EC
- **APRIL 15, 2026:** Final FC approved
- **MAY 12, 2026:** Amended EC approved
- **MAY 13, 2026:** Exempted from Wildlife clearance and cleared for handover of forest land

It received forest clearance on April 15 and environment clearance on May 12, a day before it was exempted from obtaining wildlife clearance.

According to the project site map submitted by Lloyds to the Ministry of Environment, Forest and Climate Change, the area falls in forest compartments 196, 197, 273, 274, 275, 276, 298, 300 and 301. All but one (300) of these compartments are identified as part of the Tadoba-Indravati tiger corridor in the National Tiger Conservation Authority (NTCA)-approved Tiger Conservation Plan (TCP) of the Tadoba Andhari tiger reserve.

Tiger corridors are vital wildlife pathways that connect tiger habitats, enabling animal movement, gene flow, and long-term survival. Under the Wildlife Act, development projects involving land in or around tiger reserves or corridors require statutory clearance from the standing committee of the National Board for Wildlife. A TCP is a 10-year roadmap for every tiger reserve approved by NTCA.

Under 'Category of Area Required', Lloyds itself identified the project site as a "Tiger Corridor" in its wildlife clearance application submitted in March 2024.

Yet, records show that on May 13, Maharashtra Chief Wildlife Warden (CWLW) M Srinivasa Reddy exempted the project from obtaining wildlife clearance and recommended handover of forest land to the company, stating that the project area was not within or near any national park, sanctuary, tiger reserve or corridor identified in any TCP.

When contacted, Reddy said: "That information was provided by a senior officer of the rank of conservator of forests. Now I will have to check if it is incorrect."

As reported by *The Indian Express*, the issue of tiger corridors became contentious when the NTCA did a volte-face in August 2025. Within a month of telling the Bombay High Court that tiger corridor identification should consider multiple scientific studies and parameters, it narrowed the scope to only 32 "least cost pathways" identified in 2014 and corridors listed in approved TCPs.

On March 9 this year, records show, the nodal officer of the Maharashtra forest department sought further relaxation, writing to the MoEF that only 32 forest links "should be treated as recognised legal corridors" and that inclusion of corridors identified in TCPs "may create ambiguity."

On April 24, Reddy cautioned that the nodal officer's contention "does not appear to be legally correct." In a letter to the Additional Chief Secretary (Forests), he pointed out corridors identified in TCPs by the NTCA under the Wildlife (Protection) Act, 1972, "carry statutory relevance" for scrutiny of project proposals.

Reddy further requested the state government to "issue suitable clarification" that tiger corridor scrutiny in the state should include those identified in TCPs, as well as "scientifically justified additional corridors… to ensure legally sound scrutiny of diversion proposals."

In less than a month, Reddy withdrew the letter on May 21.

Asked if there was a rethink on factoring in corridors identified in TCPs, Reddy said the question should be addressed to the NTCA.

When contacted, NTCA member secretary Sanjay Kumar said: "Like tiger reserve buffers, corridors identified in approved TCPs are meant for ecologically compatible land uses which generally exclude activities such as mining and industries."

Last July, the NTCA told the Bombay High Court that it was "in the process of the refinement of tiger corridors" based on the all-India tiger estimation data. That process remains incomplete, even as decisions on project clearances continue to hinge on how tiger corridors are defined.

Herbal cigarettes as harmful as tobacco cigarettes: Study

Ritu Sharma
Ahmedabad, May 31

HERBAL CIGARETTES, which are outside India's tobacco control laws, can be as damaging as tobacco cigarettes, a joint study by Indian Institute of Technology Gandhinagar (IITGN) and University of Illinois Urbana-Champaign (UIUC), US, has found.

Widely marketed in India and abroad as "natural, tobacco-free and even therapeutic alternatives" to conventional cigarettes, herbal cigarettes produce emissions that can be comparable to — or even more damaging than — tobacco smoke, the study, released ahead of the World No Tobacco day, said.

Notably, the study compared emissions from two of India's best-selling tobacco brands and four popular herbal cigarettes containing basil, clove, cinnamon, mint, green tea, water lily and chamomile. Two herbal brands used tendu (ebony) leaf wrappers, identical to those used in bidis, India's most widely consumed smoking product.

The research, titled 'The lure of 'Healthier Smoke': Comparative physical, chemical, and oxidative potential characterization of emissions from herbal and tobacco cigarettes', was published last week in the 'Journal of Hazardous Materials'. The research paper was co-authored by Alok Kumar Thakur and Sameer Patel from IIT Gandhinagar and P S Ganesh Subramanian and Vishal Verma from the University of Illinois, Urbana-Champaign, US.

"The research paper presents a comprehensive comparison of the physical, chemical, and oxidative properties of smoke from commercially available herbal and tobacco cigarettes in the Indian market," co-author Prof Sameer Patel, Assistant Professor, Department of Civil Engineering and Chemical Engineering, IITGN, told *The Indian Express*.

Delimitation Bill

eyeing a political realignment in Parliament. This, even as the main Opposition Congress has categorically said that the Government should follow due process and consult all parties before taking any step.

Sources said the Union Home Ministry is preparing a fresh delimitation Bill after its attempt to pass the Constitution (One Hundred and Thirty First Amendment) Bill, 2026 fell short of the required two-thirds majority in Lok Sabha in April.

The renewed push follows the April-May Assembly elections, which dealt a severe blow to the TMC and DMK in West Bengal and Tamil Nadu, disrupting Opposition dynamics.

Sources said that within the DMK, following its shock defeat and the Congress's move to join the TVK government, there are voices open to negotiating "on specific issues" with the BJP. At the same time, the BJP is closely watching the deepening divisions within the TMC and growing anger against its leadership in West Bengal. How and when these play out in Parliament is also a factor in the BJP strategy going forward.

In parallel, work is on to formalise the One Nation One Election Bill that seeks to synchronise Lok Sabha and Assembly polls. Currently, it is being reviewed by a 39-member Joint Parliamentary Committee (JPC). Asked about the proposed legislation, JPC chairman P P Chaudhary told *The Indian Express:* "The law will be amended soon. We are making good progress as far as the report is concerned, and we will submit the report in time." The JPC's tenure has been extended until the first day of the last week of the Monsoon session.

Another BJP leader said this law's rollout could be phased, given the varying terms of state assemblies. "For example, five states had elections just now. Seven states (UP, Uttarakhand, Punjab, Goa, Manipur, Himachal Pradesh and Gujarat) have Assembly elections due in 2027. So there is a possibility of One Nation One Election being implemented in phases," he said.

The BJP is learnt to have reached out to the DMK on this. "The BJP has already approached the DMK, proposing changes to the present form of the (One Nation One Election) Bill to address the party's concerns. The draft Bill will be shared with DMK leaders to seek their support," sources said.

"Since the BJP and the DMK have been on opposite sides on a number of issues, coming together will be a gradual process. It will be issue-based at first," said a BJP leader familiar with the latest political developments in Tamil Nadu.

A senior DMK leader said: "On issues such as delimitation and One Nation One Election, the party's position has always been guided by Tamil Nadu's interests rather than ideology alone. If the Centre provides a credible assurance that states, which successfully implemented population control measures, will not be penalised in parliamentary representation, and if the existing representation ratio is protected through a mutually acceptable formula, there is no reason to reject a proposal... Our concern is safeguarding Tamil Nadu's voice in Parliament."

DMK leaders, however, described suggestions of a political arrangement with the BJP as "premature", citing ideological and political differences. "We are nowhere near discussing alliances. The DMK's differences with the BJP are well known. But politics is not conducted in absolutes. Indian parliamentary history has seen parties extend support on specific issues or provide stability under extraordinary circumstances, as happened during the Vajpayee years. Any future decision would depend entirely on the political context and whether Tamil Nadu's interests and federal rights are protected," a former DMK minister said.

When the delimitation Bill was taken up, the Opposition INDIA bloc voted as a united front with 230 MPs, including those from the Congress, DMK (22) and TMC (28). The bloc had objected to the Bill linking the contentious expansion of the Lok Sabha and the rollout of the women's reservation Act, and had raised fears about the marginalisation of southern states' representation.

During the debate, Union Home Minister Amit Shah had said the Government was willing to bring an amendment to ensure a uniform 50% increase in Lok Sabha seats across states if the Opposition agreed to support the Bill. He said that by this formula, the southern states would not lose out.

Asked whether the Congress would support a fresh delimitation Bill if its concerns were addressed, party leader Jairam Ramesh told *The Indian Express:* "First there should be an all-party meeting; give the proposals in writing — our last experience showed that there is a world of difference from what was promised and what came in the Bill. There should be discussions, and parties should get time to finalise their stances... because it's a Constitution amendment... You cannot bring in amendments during the debate."

Pointing out that the Treasury benches had held discussions with regional parties while keeping the Congress and key Opposition parties out, Ramesh said: "Even what was conveyed to them orally was not there in the draft Bill. It showed their intentions were not clear."

While the BJP leadership is in contact with DMK leaders, including former chief minister M K Stalin, the party appears to have a different plan in West Bengal. "The internal fissures in the TMC and the intensifying anger against the party's leaders have made it vulnerable. Politics in Bengal has changed. The section that is deeply disappointed with the current TMC leadership is growing stronger by the day. TMC is not like the Congress; it is a regional party, and there could be a development, including a split, that the BJP could use to its advantage in Parliament," said a senior BJP leader.

Days after her party lost power in West Bengal, TMC chairperson Mamata Banerjee had launched a scathing attack on the new BJP government, accusing it of using political pressure and intimidation against opponents. "The more the BJP tortures TMC in Bengal, the more problems it will face in New Delhi... Our party workers and elected representatives are being tortured and intimidated, but we are holding on," she said.

CBSE's OSM CHAOS

The new digital evaluation system, implemented for the first time this year for Class 12 board exams, has been under fire amid claims of tampering of answer books and rushed roll out. With students — worried over college admission deadlines — complaining of challenges related to re-evaluation times, the CBSE said that the Post-Result Activities portal becomes operational from today.

Vidheesha Kuntamalla & Sophiya Mathew
New Delhi, May 31

IT WAS not every day that 17-year-old Sachi, a student from Delhi, woke up before 5 am, trying repeatedly to open a website, while the rest of the city slept. But earlier this month, she was trying to open the Central Board of Secondary Education (CBSE) re-evaluation portal — the website had been crashing for days as students across the country tried to obtain scanned copies of their answer books after the declaration of the Class 12 board examination results on May 13.

For days, as she tried, the site would open briefly or freeze. Sachi thought the best bet to access it would be to wake up before dawn when, as she hoped, fewer students would be online. "It finally worked," she recalled, accessing her scanning answer books.

For thousands of students like Sachi, CBSE's On-Screen Marking (OSM), a digital evaluation system introduced for Class 12 board examination for the first time this year, has raised major concerns about possible errors in their result. Many have even questioned if the scanned answer books they accessed even belonged to them.

For, 17-year-old Kushal Jain getting a scanned copy of the Chemistry answer book turned into an ordeal. He got a score of 97 in Physics and Mathematics, each, but only 79 in Chemistry. "I had expected a score above 95. When the results were declared, I assumed there had been a mistake." But the re-evaluation portal, he said, repeatedly malfunctioned and dates for re-evaluation were extended by CBSE.

On May 29, CBSE ultimately announced that the Post-Result Activities portal — for verification and re-evaluation — would only become operational from June 1, saying the delay was necessary to ensure a "transparent and glitch-free process" and maintain the "highest standards and protocols of evaluation."

But students are worried as deadlines are nearing for admissions to the college of their choice. "The registration for Joint Admission Counselling (JAC), Delhi, which conducts admissions to the B Tech and B Arch programmes, closes on June 9. My counselling starts on June 2. CBSE keeps extending the dates and students are stuck in limbo," said Kushal.

"If my marks increase," he added, "my overall percentage crosses 95." But the revised score may arrive after critical decisions regarding the college admissions have already been made.

For students pursuing engineering, like 17-year-old Moksh Yadav, a handful of marks can alter eligibility, counselling prospects and admission outcomes. Many engineering institutions require candidates to secure at least 75% marks in Class 12, in addition to qualifying through entrance examinations such as Joint Entrance Examination (JEE).

However, it was no less than a rude shock for Moksh when he had opened the scanned copy of his Class 12 English answer book — his score was 33 out of 80. "The handwriting was unfamiliar, entire sections were blank and the answer book had only 12 pages," he said. He distinctly remembered using a supplementary sheet during the examination. "The first page looked like mine," he said. "After that, the entire answer book, it seemed, belonged to someone else."

While he hoped to score around 78-80%, the English score brought his overall percentage down. "I wasn't expecting perfection. I just expected fairness," he said.

Moksh hopes to pursue Bachelor of Management Studies at Delhi University and has appeared for the Common University Entrance Test (CUET). "I did not get the marks for which I studied and wrote the examination... this could alter my chances to even qualify JEE," he rued.

After CBSE responded to his re-evaluation complaint by stating that no discrepancy had been found, Moksh turned to social media. Eventually, a legal notice was sent to CBSE. "If the matter remains unresolved, my family is considering approaching the Supreme Court."

Rollout rushed, say evaluators

Even as the CBSE had promised "minimum human errors" with the launch of the new evaluation system, several principals, evaluators and senior teachers who spoke to *The Indian Express* claimed that the rollout was rushed, inadequately tested and difficult to navigate for many teachers.

Minutes of a meeting held last June of the CBSE Governing Body — the Board's highest decision-making body — show that members had recommended the implementation of the OSM system across all subjects only after pilot projects had been conducted in selected subjects and regional offices. According to teachers, this never materialised.

Instead, teachers and evaluators claimed, the dry run was held across five schools involving around 100 teachers in January this year. This, even as the CBSE claimed, that they had been preparing for nearly a year.

Formal onboarding and practice sessions began around mid-February, evaluators said. "A six-month trial run should have taken place," said a headmistress of a private school in South Delhi.

> ***Teachers said the challenges extended far beyond basic computer literacy. Many found it difficult to meet the daily targets.***

The system frequently struggled under its own weight, said some teachers. "Evaluators were forced to make sense of blurred and poorly scanned answer sheets. At times, the system even logged evaluators out, and when they logged back in, the sheet appeared as already checked," alleged a teacher, who participated in the evaluation process

According to school principals, teachers were repeatedly called in to clear backlogs as pending answer books piled up on the portal, particularly in subjects such as Physics.

The transition to a new system was especially daunting for many elderly teachers, the headmistress said, adding, "Some even admitted they did not know how to use a mouse properly."

Teachers said the challenges extended far beyond basic computer literacy. Initially, CBSE required evaluators to assess 20 answer books a day. Later, this daily limit was increased to 25.

Many evaluators found those targets difficult to meet. "I was not able to correct more than 10 sheets," one evaluator said. "Many teachers faced the same situation."

Another teacher described the physical strain of the process. "Teachers were irritated, fatigued and under pressure. They had to sit in front of screens for long hours. For elderly teachers, this can take a physical toll on their bodies."

Several evaluators claimed that some scripts were blurred, poorly scanned or difficult to decipher because of overwritten text and faint handwriting.

Since the answer book appeared as a continuous digital booklet, evaluators often felt they had little choice but to continue assessing it even when the quality was poor. "Once you start the booklet, you have to finish it," one teacher said, adding, "The pressure to meet the minimum requirements often forced evaluators to compromise and correct the paper fully anyway."

A principal of private school recalled teachers repeatedly struggling to access the platform itself. "The evaluators themselves struggled with credentials and OTPs, often unable to even access the system properly," the principal said.

What CBSE says

As the CBSE continues to be under fire, a fresh statement was put out on Sunday. "We have been closely monitoring the vulnerabilities in the OnMark portal of our service provider that are being flagged in the public domain. An expert team of cybersecurity professionals has been deployed over the last few days from across various arms of the government as well as the IITs to fortify these systems, including taking them over to a more secure set up. The identified vulnerabilities have been contained, and other exploitable weaknesses are being ruled out."

Earlier, officials had told *The Indian Express* that the Board received 2.94 lakh applications seeking copies of 8.56 lakh answer books this year — more than double last year's 1.31 lakh applications for 2.82 lakh answer books. Officials and educators attributed part of the surge to a decline in top scores, with the number of students scoring above 90% falling by around 16% compared to 2025.

The CBSE is also considering imposing a financial penalty on Hyderabad-based vendor Coempt Edutech Pvt Ltd handling the OSM system, *The Indian Express* reported on May 29. Sources said the CBSE has identified 5,000 blurred answer book scans so far and 23 cases where students received scanned copies belonging to another candidate's answer sheet.

However, officials have defended the digital evaluation process, saying it involves three separate layers of quality checks before answer books reach evaluators. Scripts were reviewed during scanning, reviewed again during uploading and evaluators themselves could reject illegible scans.

In the case of mismatched answer sheets, CBSE pointed to the anonymisation stage, suggesting that errors may have occurred while handling the "flying slips" used to conceal student identities before evaluation.

Drop in pass percentage

CLASS 12 pass percentage dropped from 88.39% last year to 85.29% this year

A LOWER pass percentage than this year was in 2019 at 83.40%.

• The 12-year road to OSM

Early experiment: 2013-14

CBSE first experimented with On-Screen Marking (OSM) in 2013-14 for select subjects and regional offices, working with technology vendors including TCS. While some centres reported satisfactory outcomes, operational and infrastructural challenges prevented the system from being expanded nationwide at the time.

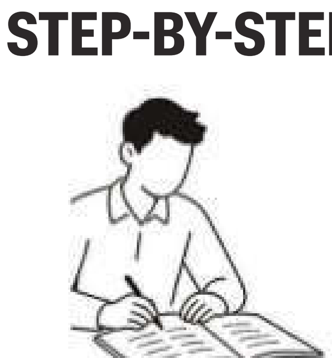
Proposed pilot: 2024-25

CBSE proposed that OSM be implemented from the 2024-25 exams in "small-volume subjects" (those with fewer examinees). Targeted for the pilot were a few subjects in the main examination for both Classes 10 and 12, and the science or mathematics supplementary exams. Class 12 Applied Mathematics was specifically flagged for OSM review.

Controversial implementation: 2026

The system has been used by examination boards like Cambridge (UK), IB, and Pearson for years. CBSE decided to introduce OSM for the evaluation of all Class 12 answer books beginning with the 2026 exams, covering all 116 subjects at full scale. The rollout has run into serious issues, with allegations of marking discrepancies, blurred scans, and technical glitches in the portal.

• STEP-BY-STEP



1

WRITING

- Students appeared for exams at designated CBSE examination centres.
- Answer books were sent to CBSE's regional offices, where they went through a "secrecy" process to conceal identifying details to make sure that evaluation was anonymous.

2

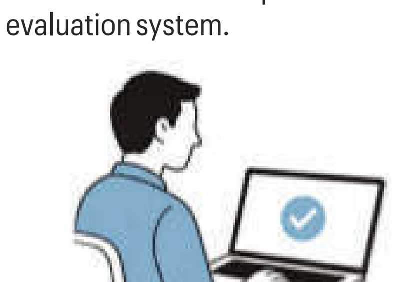
SCANNING

- Each answer booklet was digitally scanned at the regional office.
- Officials were supposed to check whether the scanned pages were clear, complete, and properly uploaded.
- After a second verification, the digitised answer sheets were uploaded to CBSE's OSM evaluation system.

3

MARKING

- Examiners login credentials; CBSE uploaded marking schemes for all subjects.
- Evaluators read scanned answer sheets online, awarded marks into the digital system.
- Software automatically added the marks to reduce calculation errors.



4

SUBMISSION

- Some evaluated answer sheets were supposed to be randomly reviewed by senior evaluators, including Additional Head Examiners and Head Examiners.
- After evaluation and checking, the digitally marked answer books were submitted into the system.

• OSM challenged: Allahabad HC, May 2026

CBSE's OSM for Class 12 has been challenged before the Lucknow Bench of the Allahabad High Court. The PIL alleged that the rushed implementation of the mechanism has caused "systemic institutional failure" in the assessment of students.

Re-evaluation scrapped — Delhi HC (2017)

CBSE had abolished the re-evaluation system for Board exam answer sheets. When the matter came before the Delhi HC, the court was shown answer sheets where completely correct responses had been awarded zero marks. The bench asked CBSE why it had done away with re-evaluation altogether.

COVID evaluation controversy — SC (2021)

After Class 12 Board exams were cancelled due to Covid-19, CBSE adopted a 30:30:40 formula to calculate students' marks based on their performance in classes 10, 11, and 12 respectively. A group of pass-out students petitioned the Supreme Court alleging that their marks had not been computed using this formula. They also alleged that CBSE's dispute resolution mechanism for aggrieved students existed only on paper.

Class 10 Internal Assessment — Delhi HC (2021)

CBSE announced that Class 10 students would be assessed entirely on the basis of internal marks awarded by their schools. An NGO challenged this before Delhi HC, arguing that it created serious risks of manipulation and exploitation of students and parents by schools. The court issued notice to the Central government and the Delhi government, seeking their response to the petition.

Key red flags: the process and debate

Vidheesha Kuntamalla
New Delhi, May 31

The OSM process

Rather than marking the paper answer books submitted by examinees, teachers assess scanned copies of the answer books that have been anonymised and put on a secure digital platform. CBSE adopted OSM this year to make evaluation more uniform, efficient, and secure with less regional variations in marking, no totalling mistakes, and better monitoring of evaluators.

"However, the attempt to speed up the marking process ignores the importance of discussions and collective understanding among teachers that helps them grow in the job," Dr Latika Gupta, assistant professor of education at Delhi University, said.

Not flawed, but...

A study commissioned by Britain's examinations regulator found that "markers using online marking systems showed greater consistency in the application of mark schemes than those using traditional paper-based methods". Assessment researcher Victoria Crisp has argued that digital systems help in standardisation and quality monitoring.

However, Cambridge Assessment researchers have noted that problems relating to scanning quality, software performance and implementation can still affect outcomes.

Edward Vickers, professor of comparative education at Fukuoka University in Japan, said: "The real question with the Indian system and this current scandal is maybe not the system or the idea behind it, but the practicalities of whether the system was trialled and piloted." According to Vickers, "What seems to have happened suggests that the system wasn't ready, the CBSE itself wasn't ready, the teachers weren't ready."

Outside India

In the UK, the Assessment and Qualification Alliance , Oxford Cambridge and RSA Examination and Pearson Edexcel have used online marking for many years. According to Britain's regulator Ofqual, online marking was introduced to improve quality control, increase efficiency, and strengthen examiner monitoring. The International Baccalaureate (IB), whose examinations are taken in more than 150 countries, also relies heavily on digital evaluation.

But Vickers pointed out that Britain's General Certificate of Secondary Education (GCSE) and A-levels were different because they frequently included long-form written responses and essays. "Technology can help make the process more efficient. But we need to be very careful about seeing technology as taking over the role of actually assessing students' answers... The human markers have to be in charge," Vickers said. **LONGER VERSION WWW.INDIANEXPRESS.COM**

New CDS General N S Raja Subramani in New Delhi on Sunday. PTI

Gen Subramani takes over as CDS, prioritises tri-services synergy

Press Trust of India
New Delhi, May 31

GENERAL N S Raja Subramani, widely known as an expert on Pakistan and China, on Sunday took charge as India's new Chief of Defence Staff with a primary mandate to implement the ambitious military theaterisation plan and bolster tri-services synergy. He succeeded Gen Anil Chauhan, who signed off after completing his tenure as the country's senior-most military commander on Saturday.

Gen Subramani was serving as the Military Advisor at the National Security Council Secretariat (NSCS). He retired as the Vice Chief of Army Staff on July 31 last year. The military officer, shortly after taking charge, said transformation of the armed forces and organisational reforms to enhance tri-services synergy and integration will be his primary focus.

"We will accelerate the development, induction and integration of indigenous weapons in our armed forces," he said in a brief media statement.

"Our armed forces consistently demonstrated professionalism and operational decisiveness in safeguarding our national interests. We are committed to protecting the sovereignty and territorial integrity of our country," he said.

"I assure citizens of India that armed forces will continue to serve the nation with dedication, courage, honour and professionalism," he said. Gen Subramani noted that he would work towards accelerating the development, induction and integration of indigenous weapon systems into the armed forces.

Calling for "greater collaboration" between the military, industry and academia, the new CDS said the Indian Army, the Navy, Air Force, the defence ministry and the strategic institutions stand united as part of the "whole of nation approach" to strengthen India's security.

Adm Swaminathan new Navy chief

Press Trust of India
New Delhi, May 31

ADMIRAL KRISHNA Swaminathan on Sunday assumed charge as India's 27th Chief of the Naval Staff and immediately prioritised peak operational readiness as his top focus to navigate a fast-evolving regional maritime security landscape. He succeeds Admiral Dinesh Kumar Tripathi, who superannuated after 41 years of distinguished service.

ANI
Admiral Krishna Swaminathan

Admiral Swaminathan was earlier serving as the Flag Officer Commanding-in-Chief of the Western Naval Command.

The new Navy Chief described the regional security environment as "challenging, complex and unpredictable", and asserted that maintaining top-level operational readiness will be his highest priority.

"The Indian Navy stands vigilant to protect national interests…and is very actively deployed in a regional security environment that continues to remain challenging, complex, unpredictable, and uncertain," he told reporters.

'RIGHT OF SEX WORKERS WITHOUT RIGHT TO SEX WORK'

SC asks govt to make law exempting trafficked sex workers from prosecution

Ananthakrishnan G
New Delhi, May 31

UNDERLINING THAT there can't be a one-size-fits-all approach in the application of the Immoral Trafficking Prevention Act (ITPA) to tackle commercial sexual exploitation (CSE), the Supreme Court has laid out an elaborate 'Victim Protection Plan', securing the rights of "voluntary" sex workers, and called upon the government to consider amending the law to exempt trafficked victims from being prosecuted for prostitution. It also sought that the new law protects the victims from "sexual abuse" by police personnel when in custody.

A bench of Justices J B Pardiwala and R Mahadevan, which exercised powers under Article 142 of the Constitution, said in its May 29 judgment that the plan which details "the manner in which preventive, protective and rehabilitative measures are to be taken to safeguard the fundamental rights of the victims of trafficking for CSE...would hold the field until the Parliament enacts a law on the subject". The court noted that "rights of sex workers can exist without there being a right to sex work".

The 297-page judgment sought to distinguish those trafficked for prostitution from adults who indulge in it "voluntarily" and said that "by conflating prostitution and trafficking", the Act as it stands processes "those trafficked against their will...those who were trafficked but continue voluntarily..." and "those who have chosen sex work for themselves...through the same mechanism under Section 17, without differentiation."

It said that "ITPA's failure to accord rights and protections to" the latter "combined with the deep social stigma attached to sex work, results in voluntary adult sex workers being isolated, marginalised, and unable to access protections that the law extends to them".

Writing for the bench, Justice Pardiwala said, "The larger society's view of sex work is deeply rooted in moral judgment... This attitude is perhaps most starkly reflected in the deeply derogatory terminology that continues to be used in common parlance in Hindi and English to describe voluntary adult sex workers. The consequences of this stigma are severe and far-reaching."

The new plan, with an emphasis on dignity and human rights, says, "Victims shall not, at any stage, be treated as criminals or subjected to measures associated with criminal liability."

The court said that "for victims trafficked for CSE, the crime violates their right to live with dignity directly, i.e., it commodifies, impoverishes, and stigmatises. But the absence of dignity also precedes the crime, i.e., lack of access to minimum material necessities, is predominantly what makes victims vulnerable to traffickers in the first place."

A one-size-fits-all approach, applied to the heterogeneous group of 'victims of trafficking for CSE', is unlikely to serve their interests properly. In fact, it might lead to unintended consequences.

The court laid down that when sex workers are produced before a Magistrate, he or she "must conduct an initial threshold inquiry into whether the person before him is a voluntary adult sex worker who does not wish to be subjected to long term safe custody".

"Where the victim clearly and unequivocally states that she is engaging in prostitution voluntarily and does not wish to be placed in long-term safe custody...the Magistrate shall pass an order directing a preliminary inquiry into whether her engagement in prostitution is voluntary in nature," the court said.

Explaining why the magistrates should have the option of placing the victim under custody upon investigation, the bench said, "It is a known fact that traffickers exercise deep and pervasive control over their victims through...physical force, psychological manipulation, coercion, and sustained intimidation," adding that the victim's consent may be exploited in such cases.

The court pointed out that "the legal infrastructure addressing human trafficking for the purposes of CSE in India is broad but fragmented, comprising multiple statutory instruments" and called upon the government to come out with a "separate and comprehensive legislation on trafficking, one that addresses not only trafficking for CSE but all other forms of exploitation".

It referred to the "continued prevalence of instances of abuse and reports indicating the complicity of police officers in the crime of trafficking" and urged the government to create a specific offence where police officers "compel" or "seduce" a victim in their charge or custody to "illicit sexual intercourse", and a rebuttable presumption that a police officer who "delays" the production of a rescued victim before a magistrate has "wrongfully confined her".

The court also laid down elaborate procedures for police raids to rescue victims including that "no person shall be abused either verbally or physically" and "the manner in which the persons are rescued must not involve elements of unnecessary physical force or manhandling".

The court further added, "The members of the rescue team shall endeavour...to identify whether the persons intended to be rescued are voluntary sex workers or not...

"It must be ensured that while photography and videography of the rescue site is done, the victims are excluded."

'Encroached India land': Nepal PM triggers a row

Shirish B Pradhan
Kathmandu, May 31

NEPAL PRIME Minister Balendra Shah on Sunday said he has learnt about his country "encroaching" territories in India, as he sought to answer questions in parliament on a long-running border dispute and instead triggered a controversy.

In his maiden appearance in the parliament session that started May 11, Shah also said India and Nepal have agreed to take the help of historians, surveyors and experts to seek a resolution, adding that Kathmandu has also taken up the matter with China and the UK.

Hours later, Nepal Foreign Ministry said the PM's comments were related to "no-man's land encroachments" and "cross-border occupation" between the two countries, rather than any territorial claims.

Nepal and India have had an old boundary dispute over Lipulekh, Limpiyadhura and Kalapani, with both countries claiming the areas. India maintains that the territories are part of Uttarakhand and has said

Nepal PM Balendra Shah

the issue should be addressed through bilateral dialogue.

There was no immediate response from India on Shah's comments but earlier this month, while rejecting Nepal's objection to the Kailash Mansarovar Yatra through the long-established Lipulekh Pass, it had dismissed Kathmandu's territorial claims over the region as an "unilateral artificial enlargement" that New Delhi finds "untenable".

On Sunday, Shah said: "The Nepal government has officially sent a diplomatic note to India, mentioning the issue of encroachment of territories by India, including Lipulekh, and we have already received their response. Both the countries have agreed to resolve the issue sitting together with the help of historians, surveyors and concerned experts through diplomatic means." When specifically asked by a lawmaker about the government's view on the dispute concerning the Limpiyadhura, Lipulekh and Kalapani areas, Shah said: "You will be surprised to know about a fact, which I learnt recently, only after becoming the PM. It is not only India that has encroached Nepalese territories, but Nepal has also encroached India's territories in many places. Now both countries should study facts and sit together as friends and resolve the issue."

The three places are located near the trijunction of India, Tibet and Nepal. Shah said Kathmandu has also taken up the matter with China and the UK. He said he raised the matter with the UK because it dates back to a period when the British left the region.

His comments about Nepal encroaching Indian territories triggered a controversy with opposition lawmakers, including Basana Thapa of the Nepali Congress and Ramesh Malla of the Nepali Communist Party, demanding that they be expunged from the parliamentary record. They said the PM should either provide evidence to support his claim or withdraw the statement. Former Nepal Ambassador to India Nilambara Acharya told Kantipuronline media portal that Shah has "no information regarding Indian territories being encroached by Nepal."

Asserting that Nepal's current border is based on the Sugauli Treaty of 1816, Nepal's Foreign Office said, "Limpiyadhura, Lipulekh and Kalapani and Susta in Nepal-India border area are territories that remain to be demarcated."

"There are problems relating to cross-border occupation and no-man's land (Dasgaja) encroachments in some other areas besides these. What the Prime Minister mentioned in Parliament was related mainly with Dasgaja encroachment and cross-border occupation," the Foreign Ministry spokesperson said in a statement. **PTI**

THE INDIAN EXPRESS

The Editorial Page

MONDAY, JUNE 1, 2026

Austerity is needed, but energy security requires breaking silos in government

OVER THE BARREL

BY VIKRAM S MEHTA

THE PM has issued a clarion call for energy austerity. This is timely because even when the Straits of Hormuz reopen, India will continue to face elevated energy prices and supply uncertainty. Worldwide, stocks of crude oil and petroleum products have been depleted. These will have to be replenished. Eight refineries in the Persian Gulf and Qatar's Ras Laffan LNG terminal have been damaged.

This will keep the petroleum market tight. In addition, while a "no war-no peace" scenario (like the relationship between North Korea and South Korea, or the current ties between India and Pakistan) may continue indefinitely, India cannot discount the possibility of miscalculation, misadventure, and/or misunderstanding triggering a violent rupture.

The PM's call does, however, raise several questions. Who, other than the PM, has the executive mandate within our administrative and institutional set-up to put flesh on his directive? Who will define the targets to be achieved? Who will highlight the cross-cutting linkages within the energy sector (petroleum, coal, renewables, power, nuclear) and between energy and other economic sectors (fertiliser, food, water, environment, semiconductors, heavy industry) to ensure a "whole-of-system" approach to the management, monitoring, and evaluation of energy policy and security?

The answer is no one. There is no individual or institution with executive authority over energy. Some years ago, I proposed that the government revert to the *status quo ante* of 1992 and establish a Ministry of Energy, with the Minister ranked alongside the Ministers of Defence, Home, Finance, and External Affairs as one of the senior-most members of the cabinet. I suggested that the current cabinet-level ministries of petroleum, coal, renewables, and power be subsumed under this ministry.

This suggestion had no takers, possibly because it was politically infeasible. With 23 political parties in the coalition government, all of whom had to be accommodated, a proposal that reduced the size of the cabinet was never likely to gain traction. Today, this is no longer a constraint. The proposal could therefore be brought back to the table. I would suggest that NITI Aayog be directed to lead a discussion on the subject.

That said, I recognise that administrative and institutional change is never easy. Powerful vested interests must be navigated. A second-best option would, therefore, be for the PM to create a "Department of Energy Resources and Security" (DERS) within the Prime Minister's Office, headed by a cabinet-ranked official and staffed by a multidisciplinary cadre of skilled specialists. The purpose would be to supplement, not duplicate or dilute, the responsibilities of the existing energy-related ministries. They would continue "as is". There would be no change to the present administrative structure.

Specifically, I would recommend that DERS be vested with executive responsibility for the following five functions.

First, to provide a platform for a "whole-of-system" integrated approach to energy policy planning, implementation, monitoring, and evaluation (*à la* the Chief of Defence Staff). Most importantly, it should identify, study, and evaluate intra-energy and energy-non-energy economic, social, and environmental linkages to ensure efficiency of use, supply resilience, competitiveness, and decarbonisation.

Second, to ensure symmetry of investment across the energy value chain (generation, transmission, storage, distribution). Currently, this requires coordination among multiple ministries at the Centre and between the Centre and states. Lack of symmetry results in avoidable economic loss. According to economist Swaminathan Aiyar, 4 GW of solar energy is stranded in Rajasthan alone because of inadequate evacuation capability. Third, to provide a platform for "India Energy Inc." to leverage its market power to secure energy at competitive prices, attract risk capital for the development of indigenous energy resources, and forge multilateral and bilateral research and technology partnerships.

Fourth, to serve as the "Energy Ombudsman". Currently, there are a plethora of energy regulators. All have well-defined roles and responsibilities, but there is overlap and, on occasion, contention. The ombudsman should streamline and standardise energy regulation, particularly in relation to strategic reserves, inter-state wheeling of power, EV charging infrastructure, battery technology, time-of-day pricing, and dispute resolution. It should create a database of energy regulations worldwide, establish a forum for information exchange and best practices, and supervise the development of regulatory talent.

> The government should consider enacting an 'Energy Responsibility and Security Act' that defines the roles and responsibilities of every citizen in securing energy 'atmanirbharta' and sustainability

Finally, DERS should serve as the nodal communication point for energy-related matters. Energy is a politically sensitive subject. The PM currently bears the mantle of educating the public on the macroeconomic and cross-cutting ramifications of energy. This responsibility should devolve to DERS.

In this context, the government should consider enacting an "Energy Responsibility and Security Act" that defines the roles and responsibilities of every citizen in securing energy *atmanirbharta* and sustainability, while laying out a roadmap towards these objectives. DERS should take the lead in preparing the first draft.

The ripple effects of the closure of the Straits of Hormuz bring to mind the response economists at the London School of Economics gave Queen Elizabeth when she asked how such a conglomeration of brilliant people had failed to anticipate the economic and financial crisis of 2008. They explained that while "everyone seemed to be doing their own job properly" and "on its own merit", there was "no one who saw the whole picture" and appreciated that "while individual risks may rightly have been small", the collective impact of these risks could trigger a systemic collapse.

This answer mirrors the problem of energy policy in India. Those involved with energy appreciate the nature of our energy challenge. They know what must be done. But because each of them has a mandate over only one component of energy policy, they lack a line of sight on the systemic consequences of seemingly "small" failures. This must change. We need to break silos and create a system that enables executive oversight over the totality of energy policy and security.

The writer is chairman and distinguished fellow of the Centre for Social and Economic Progress

Science collaboration is space compass, not conflict

LAST WEEK, NASA announced an ambitious roadmap for sustained operations on the Moon over the next decade. The agency reportedly envisions creating a base on the lunar South Pole in three parts: Getting to the moon's surface and beginning experiments from 2026 to 2029; "initial operations", including potential nuclear power installations in the next three years; and commencing work on a "semi-permanent" human station after 2032. The choice of location is significant. Unlike many regions of the Moon, where periods of sunlight and darkness alternate about every two weeks, parts of the South Pole experience shorter periods of shadow. They can thus provide solar power to drive the base's energy operations. At the same time, scientists believe that ice deposits in the region's permanently shadowed parts contain records of how water and other materials moved through the solar system. Studying these samples could help understand processes that created conditions for life.

The lunar outpost is also expected to be a staging ground for expeditions to Mars. According to NASA, it could provide the first "long-term opportunity to study human physiological adaptation in a reduced-gravity field before a deep space voyage". This endeavour is closely tied to geopolitics. Over the past decade, China has built a robust space programme that has landed rovers on the Moon and brought lunar samples to its laboratories. Beijing and Moscow have also announced plans to build a nuclear-powered international space station within 100 km of the lunar South Pole. In other words, unlike the largely symbolic space competition of the last century between America and the Soviet Union, the race to the Moon and beyond is about gaining early access to strategically significant regions outside the Earth.

NASA's plans assign a prominent role to Amazon founder Jeff Bezos's aerospace company, Blue Origin, which has been tasked to develop heavy-lift rockets, lunar landers and early infrastructure for the project. Although Elon Musk's SpaceX doesn't seem to have found a place in the programme announced last week, its partnership with NASA remains central to America's space ambitions. The convergence of great-power rivalry and commercial interests could accelerate scientific progress. At the same time, there are concerns that the Cold War-era legal framework is ill-equipped to govern the changing dimensions of space exploration. There is, of course, little prospect of an imminent confrontation between the US-led Artemis programme and the Sino-Russian International Lunar Research Station. However, the emergence of competing legal regimes risks fragmenting scientific cooperation at a time when challenges such as climate change demand collaboration between nations. Mechanisms must be found to ensure that scientific advancement, rather than geopolitical competition, remains the principal driver of space exploration.

Amid rising inflation, RBI cannot neglect growth

THIS WEEK, the Reserve Bank of India's Monetary Policy Committee will deliberate on fine-tuning its stance on the repo rate -- the interest rate at which India's central bank lends money to commercial banks. Typically, when prices threaten to rise at a rate faster than what the RBI finds prudent, the central bank raises the repo rate to contain inflation. On the face of it, India is facing such a phase, with the war in Iran and the blockade of the Strait of Hormuz keeping the prices of crude oil and other critical imports high. Not surprisingly, inflation has started rising quickly. Wholesale inflation has already crossed the 8 per cent mark, and it's only a matter of time before its effects filter through as higher retail inflation for consumers. Retail inflation has already spiked to 3.5 per cent after the RBI had brought it down to 2 per cent in the last financial year. Moreover, a weak monsoon could affect the farm sector and raise food prices. Most economists outside the RBI and the government expect retail inflation to rise further to 5 per cent and stay at that level for the rest of the year. Given that the RBI targets a retail inflation rate of 4 per cent, many do not rule out a repo rate hike.

However, the RBI's choice may not be as straightforward as it seems. For one, the current rise in prices stems from a lack of supply of crude oil and other products. In the short term, there is little that the RBI can do to boost crude oil supplies. Raising interest rates may dampen India's growth momentum at a time when it needs a more conducive environment. A US-Iran peace agreement, resulting in a swift boost to oil supplies and a quick moderation in oil prices, cannot also be ruled out completely. Also, at 5 per cent, inflation would still be within the RBI's comfort band of 2 to 6 per cent.

The more important issues to watch this week will be the RBI Governor's commentary on the state of the economy. Two issues will be of particular interest. One, the rupee's exchange rate. Does the RBI want to prioritise maintaining an adequate amount of forex reserves and let the rupee find its own level, or is it willing to draw down these reserves to ensure that the rupee doesn't slide beyond a certain point? Two, what needs to be done for India to attract foreign capital. The RBI's stance will be keenly watched.

Interpreting Harappa, the Shereen Ratnagar way

JAYA MENON

SHEREEN RATNAGAR, who passed away recently, was known for her research centred on the Harappan civilisation. It is not surprising that she chose to work on the trading encounters between West Asia and the Indian Subcontinent in the third millennium BCE, given her training in Archaeology at Deccan College, Pune and the Institute of Archaeology, London, where she specialised in Mesopotamian archaeology. This became the object of her doctoral research at JNU. The book that emanated from her research (*Encounters: The Westerly Trade of the Harappa Civilisation*) was not a mere detailing of goods and objects that were moved between the Harappan region and areas to the west such as Mesopotamia, Bahrain and Oman; rather, the book, and particularly its second edition (*Trading Encounters: From the Euphrates to the Indus in the Bronze Age*), provided a nuanced interpretation of these transactions. In an interview on *harappa.com*, Shereen writes that her focus on the socio-economic and political, in contrast to what other archaeologists at the time in India were writing, was wrought by her training at JNU. It is this focus, towards archaeological context and away from solely concentrating on the object, that lies at the heart of Shereen's work.

Archaeology in India, till about the 1980s, had largely remained at the documentation stage, in a more cultural-historical frame, with the focus on "filling the gaps" and recovering occupation sequences at various sites. In contrast, Shereen's lifelong work to make sense of what had been, and was being, archaeologically recovered from Harappan sites, pointed to new ways of understanding the data. Her way was to lean heavily on the theoretical and the anthropological, which is interesting given the colonial roots of Archaeology and the strong link the discipline has always had in the Subcontinent with History. In that sense, Shereen's most significant work is the less well-known *Enquiries into the Political Organisation of Harappan Society*, where she explored the nature of Harappan polity. Her thesis in this book, on a topic that remains contentious, is a riveting one. Shereen's third book, *The End of the Great Harappan Tradition*, was an attempt to parse the available data on this question and critiqued the standard explanations of Harappan "decline" prevailing at that time.

> Shereen's work to make sense of what had been, and was being, archaeologically recovered from Harappan sites, pointed to new ways of understanding the data. Her way was to lean heavily on the theoretical and the anthropological

Shereen also engaged in more public-facing writing. Understanding Harappa was a small but informative book intent on making the Harappan civilisation accessible to most readers. One of her later works, *Makers and Shapers*, looked at everyday technologies like house construction to bring out the labour dimension, and especially the implications for production that took place within households.

Shereen's uncompromising courage was apparent in her intervention in the Babri Masjid dispute. She joined Professor Dhaneshwar Mandal to dissect a photograph from B B Lal's 1970s excavation at the Babri Masjid, which aimed to show what were called "pillar bases". The stratigraphic evidence, which they closely studied, showed that none of these "pillar bases" belonged to a single stratum and, hence, they could not have been part of a single structure. Mandal and Ratnagar were among the few archaeologists to visit the Babri Masjid site and raise questions about the 2003 excavations. In her own words: "No social science proceeds in an ideological vacuum... Being 'apolitical', in turn, often amounts to an acceptance of the *status quo*." Her words continue to be relevant considering the times through which we are currently living.

Shereen's no-nonsense attitude often filled her students with dread, a feeling I remember all too clearly as her doctoral student. At the same time, during our discussions about chapters over a cup of coffee, it was also visibly clear when an innovative idea or thought of mine would delight her. I recall at the conference that some of us students organised for her Festschrift, the one person who was all set to actively contribute, take notes, and discuss most animatedly was Shereen! That's how she was.

The writer is a retired professor, Azim Premji University, Bhopal

Safer motherhood begins before pregnancy

BHASKAR PAL

YOUNG WOMEN in urban India today are choosing to delay starting a family in pursuit of careers and stability, yet they are surrounded by anxiety-inducing conversations about fertility and late pregnancies. In contrast, many women in rural settings conceive much earlier but often navigate pregnancy with far less access to a consistent continuum of antenatal and postnatal care. In both cases, the question arises: Are we doing enough to safeguard them by providing a full spectrum of maternal care?

A new study in *The Lancet Obstetrics, Gynaecology & Women's Health* shows India's maternal mortality ratio has fallen from 384 deaths per 100,000 live births in 2000 to just 88 in 2023 — one of the steepest declines anywhere in the world. Sustaining that momentum now requires turning our attention to a condition that shapes how safely a woman experiences pregnancy long before she reaches the labour room: Anaemia.

The NFHS-V shows that 57 per cent of women aged 15-49 are anaemic, up from 53 per cent in the previous survey. Anaemia compounds the risks of haemorrhage and infection and can contribute to both maternal deaths and pre-term labour. Iron deficiency accumulates over years. Inadequate dietary diversity and heavy menstrual blood loss deplete iron stores long before a woman becomes pregnant. This is why a life-cycle approach matters. Iron and folic acid supplementation, already embedded in national programmes, must reach women early. Oral iron remains the standard first-line therapy for mild to moderate anaemia because it is accessible, affordable, and effective when taken consistently. However, adherence is difficult, absorption is unpredictable, and for some women there simply is not enough time. A woman diagnosed with severe anaemia at 32 weeks of pregnancy cannot wait three months for her haemoglobin levels to rise through daily tablets. For her — and for women who cannot tolerate oral iron — intravenous iron, particularly a formulation called Ferric Carboxymaltose (IV-FCM), is a critical option.

> The NFHS-V shows that 57 per cent of women aged 15–49 are anaemic, up from 53 per cent in the previous survey

Postpartum haemorrhage is one of the leading causes of maternal deaths in India. There are, however, proven interventions. The E-MOTIVE bundle, tested across more than 200,000 births in resource-limited settings, reduced severe bleeding outcomes by 60 per cent.

Safer motherhood begins with better nutrition, earlier testing, timely treatment, and a health system that supports women at every stage of life. Collaborative platforms such as the ADHUNA initiative, which brings clinicians together to strengthen provider awareness and share practical tools, demonstrate what becomes possible when knowledge, training, and practice move in the same direction. That effort must now become broader, better coordinated, and sustained so that both public and private healthcare systems can help women enter pregnancy with strength.

The writer is president, Federation of Obstetrics and Gynaecological Societies of India

India-Oman pact holds promises for farms, factories and small businesses

PIYUSH GOYAL

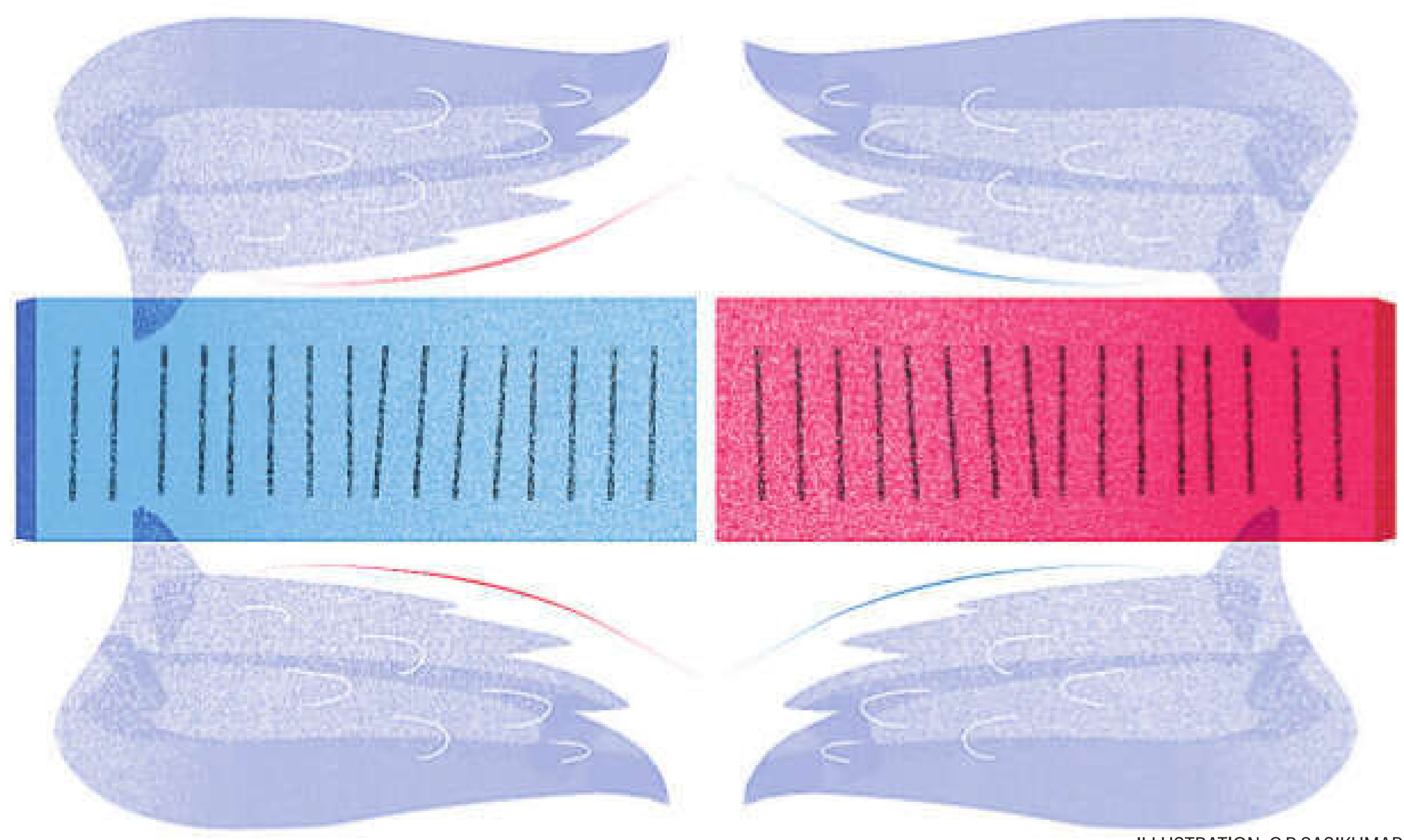


ILLUSTRATION: C R SASIKUMAR

THE INDIA–Oman Free Trade Agreement that comes into force from June 1 is a defining milestone in Prime Minister Narendra Modi's mission to create global pathways to prosperity for India's students, artisans, women, farmers, fishermen and MSMEs by opening new markets and accelerating job creation.

India has a deep economic and people-to-people connect with Oman, home to nearly 7 lakh Indians, including merchant families whose roots there go back 200–300 years. Indian remittances from Oman amount to about $2 billion annually, while more than 6,000 Indian enterprises operate there.

The Comprehensive Economic Partnership Agreement (CEPA) between the two countries significantly deepens economic and strategic ties. It immediately gives 100 percent duty-free market access in Oman to 98 per cent of tariff lines covering 99.38 per cent of exports. This marks a dramatic improvement over the pre-CEPA system that allowed only 15.3 per cent of India's exports enter Oman at zero duty. Goods from India that currently face a 5 per cent import duty in Oman, worth about $3.64 billion in exports, will become more competitive.

For India's MSMEs, the agreement could be transformative as many sectors that gain from CEPA are dominated by small businesses. Iron and steel, textiles, leather, auto components and industrial equipment are some of the sectors where MSMEs are expected to get large international orders, which will boost production, investment and hiring.

In an era of growing global instability, the CEPA offers Indian exporters a crucial opportunity to diversify markets and reduce dependence on traditional destinations facing economic slowdown and rising trade barriers.

Job creation: The trade pact benefits labour-intensive sectors such as textiles and apparel, leather & footwear, food processing, marine products, gems and jewellery, and select engineering segments, which are major employers in India.

Increased textiles exports to Oman will boost production and create jobs across major clusters such as Tirupur, Surat, Ludhiana, Panipat, Coimbatore, Karur, Bhadohi, Moradabad, Jaipur, and Ahmedabad. Artisans and weavers across India will also gain from higher international demand for their products.

Numerous jobs will also be created in major leather and footwear hubs across India, particularly in Tamil Nadu, Uttar Pradesh and West Bengal as well as parts of Maharashtra, Punjab, Karnataka and Madhya Pradesh among others.

The gems and jewellery sector is another example of how the CEPA will accelerate employment growth. India already has strong capabilities in cut and polished diamonds, gold and silver jewellery, and handmade jewellery production. With tariff barriers removed, Indian exporters will get a decisive competitive edge over European and Asian rivals. The industry estimates that exports to Oman could grow by as much as USD 150 million over the next three years. This is expected to create substantial employment opportunities in jewellery-making clusters across West Bengal, Tamil Nadu, Maharashtra, Rajasthan, and Gujarat.

> In a fragmented and protectionist world, PM Modi is sending a clear message that the new, confident India will not retreat behind walls

Farmers and fishermen: To safeguard domestic farmers and sensitive agricultural interests, India has given no tariff concession for major products such as wheat, rice, maize, millets, dairy, fruits, vegetables, edible oils, oilseeds, tea, coffee and honey.

India will get a competitive advantage over rivals in butter, honey, sweet biscuits, eggs and some confectionary products, which will increase demand for the country's farm produce and boost rural incomes.

The agreement also provides for acceptance and recognition of India's National Programme for Organic Production (NPOP) certification, which will give enormous opportunities to Indian farmers to sell organic output to Oman, which is a major food importer.

Marine products also have huge untapped potential. Oman's imports of marine products was about $119 million between 2022 and 2024. Imports from India stood at only $7.75 million, giving enormous opportunities for Indian seafood exports such as shrimp and frozen cuttlefish. The labour-intensive marine products industry is likely to generate additional jobs in fishing, processing, packaging, cold-chain logistics, and export operations.

Pharma and traditional medicine: Indian medicines approved by regulators such as USFDA, EMA, UK MHRA and TGA will receive automatic marketing authorisation in Oman within 90 days — a major win for Indian pharma exporters.

Significantly, CEPA creates opportunities for India's traditional medicine services. It provides for joint research in traditional medicine.

Services and mobility: Another important dimension of the agreement lies in services and mobility. Oman has undertaken commercially meaningful commitments in sectors of major export interest to India, including professional services, computer and IT services, education, healthcare, tourism, research and development, and environmental services. Indian professionals in areas such as accounting, engineering, medicine, construction, education, and consulting are expected to benefit from improved market access.

Oman has agreed to enhanced mobility commitments for Indian professionals and workers. Intra-corporate transferees and contractual service suppliers will be allowed stays of up to four years, while business visitors and independent professionals will receive easier temporary entry provisions. It has raised the ceiling for intra-corporate transferees from 20 per cent to 50 per cent.

The Modi government's initiatives to sign trade pacts with developed countries is part of the Prime Minister's mission to improve the life of every Indian. The agreement with Oman is a reminder that trade is a powerful instrument of growth, job creation and shared prosperity. In a fragmented and protectionist world, PM Modi is sending a clear message that the new, confident India will not retreat behind walls. It will rise through partnerships, competitiveness and global engagement.

The writer is Union Minister of Commerce and Industry, Government of India

Trump's Iran challenge — to go beyond Obama's deal

R SWAMINATHAN

THERE WAS a great sense of optimism nine days ago when Donald Trump held a summit phone call with Pakistan, Saudi Arabia, Qatar, the UAE, Bahrain, Egypt, and Turkey — and spoke separately with Israel — to announce that a US-Iran MoU was close to being finalised. However, the MoU, now labelled the Islamabad Declaration, continues to hang fire.

Under its provisional framework, Iran will reopen the Strait of Hormuz to pre-war commercial shipping levels within 30 days of signing the MoU, though restrictions will remain in place for military vessels. In return, the US will gradually lift its naval blockade, temporarily suspend oil export sanctions, and pledge comprehensive sanctions relief alongside the unfreezing of Iranian assets upon a final agreement. On the nuclear file, Iran will formally reiterate its commitment to never pursue nuclear weapons. Concessions on this issue will be linked to the US sanctions relief. Notably, the MoU does not include any reference to Iran restricting its missile capabilities or its support for regional proxies.

Trump's announcement about the interim MoU led to rare public backlash from Republicans, who saw it as a premature capitulation that leaves Iran strategically intact. This reaction was expected, given that mainstream Republican criticism of the 2015 Joint Comprehensive Plan of Action (JCPOA) has remained one of the most significant foreign policy battlegrounds of the past decade. To manage the backlash, Trump publicly instructed his negotiators not to rush the final text until they secure a better deal. He framed the MoU as a vehicle to expand the Abraham Accords, declaring it mandatory for Muslim nations involved in the Iran talks to join the Accords — notwithstanding the fact that none of these countries will be in a position to alter their policies until a solution to the Palestinian issue is accepted. The US's limited attacks on southern Iran last Monday and Thursday, along with its green-lighting of Israel's latest major military operations in Lebanon, should also be viewed in this context.

The fact that Iran has not broken off negotiations despite recent US strikes and Israel's massive offensive in Lebanon suggests that Tehran wants to avoid escalation, opting instead to maintain pressure on Washington to accommodate its interests. Iran has the advantage of time. Tehran is operating on the realisation that Trump's military threats lack follow-through. And while it seemed imminent that the MoU would be concluded soon — especially given Trump's statement that he would make his final decision following last Friday's meeting — there has been no movement forward.

> Trump is now in a worse position than at the start of the war due to the blockade of the Strait of Hormuz

Iran is insistent that the war must end on all fronts, including in Lebanon. Despite fierce opposition from Israeli PM Benjamin Netanyahu, Washington is proposing that the 60-day ceasefire extension apply to Lebanon, while explicitly preserving Israel's right to act against immediate threats from Hezbollah. However, Trump is in a weaker bargaining position with Iran than Barack Obama was in 2015. He is now in a worse position than at the start of the war due to the blockade of the Strait. Having altered the maritime status quo, Iran is now positioned to influence both the pace and the content of the peace talks.

Nevertheless, Trump must secure nuclear concessions that go far beyond the original JCPOA parameters. Unlike the sunset clauses of the JCPOA — under which critical restrictions on Iran expired after 10 to 15 years — this baseline requires Iran to accept a complete enrichment moratorium and a subsequent permanent cap. Kazakhstan, already hosting an IAEA low-enriched uranium bank, has emerged as a possible destination for Iran's 60 per cent enriched uranium. At the same time, Washington faces a major impasse over maritime sovereignty, as Iran continues to insist on long-term control of maritime passage through the Strait of Hormuz. In the end, a diplomatic breakthrough is still achievable if both parties accept compromises and secure enough narrative leverage to satisfy their domestic bases.

The writer is former Governor for India at the IAEA, Vienna, former Ambassador to Egypt and former Permanent Representative to the Arab League

The cloud is the new frontier of digital sovereignty

R S SHARMA

LAST YEAR, Nayara Energy — one of India's largest oil-refining companies — received a notice that should have alarmed every policymaker in Delhi. Microsoft informed the company that it might have to discontinue its cloud services, citing US sanctions compliance obligations arising from Nayara's Russian shareholder, Rosneft. Nayara was operating within Indian law. Its refinery is on Indian soil. But the digital infrastructure on which it ran its operations was leased from a company headquartered in Redmond, Washington — and was, therefore, subject to American law. The threat was not ultimately executed, but India had just glimpsed what digital dependence looks like when geopolitics turns hostile.

The Nayara episode is a reminder of something we have been slow to reckon with: Microsoft's cloud, Amazon's AWS, and Google's infrastructure are, in the truest sense, digital infrastructure. Hospitals run on them, banks clear transactions through them, government agencies store data on them, and entire supply chains depend on them. That dependence comes with a condition: The terms can change, and the switch can be thrown, from a boardroom in another country.

Not all digital infrastructure carries the same risk. Private digital infrastructure — the clouds, platforms, and algorithms built by global technology firms — is a commercial asset, governed by the laws of their home jurisdictions. They can be monetised, restricted, or weaponised. Then there is the digital public infrastructure (DPI), which a country builds to serve its citizens on its own terms. India's DPI stack is among the most impressive in the world: Aadhaar for identity, UPI for payments, DigiLocker and eSign for documents, ABDM for health data, and ONDC for open commerce. These systems are open, interoperable, and non-extractive. They do not mine user data for profit. Any innovator can build on them. India controls its DPI stack. What India does not fully control is the infrastructure layer on which much of this runs: The cloud, the chips, and increasingly, the AI.

If cloud dependency is today's risk, AI is tomorrow's. The LLMs now rapidly becoming central to enterprise operations and government services are controlled almost entirely by US and Chinese companies. The data they are trained on, the values they embed, the guardrails they enforce — none of these reflects Indian priorities or is subject to Indian oversight. Here lies India's central tension. Having built a world-class DPI stack at the application layer, India has allowed its dependence on foreign hyperscaler cloud — AWS, Azure, Google Cloud — to deepen substantially. The private sector, including much of the fintech and healthtech built on DPI rails, runs overwhelmingly on foreign cloud. Owning the application layer while renting the infrastructure layer is a form of digital tenancy.

None of this calls for digital isolationism. The question is not whether to engage with foreign digital infrastructure, but how to reduce strategic vulnerability to an acceptable level.

Four levers are available. First, data localisation — requiring sensitive data to reside within Indian territory — is necessary but insufficient if the cloud on which it sits remains foreign-controlled. Localisation must be accompanied by genuine operational sovereignty, including the ability of Indian authorities to audit and, in extreme conditions, take over critical infrastructure. Second, India needs a credible domestic cloud capability. MeghRaj and emerging Indian data-centre investments are a start; The goal should be ensuring that critical workloads have a sovereign fallback. Third, the DPI substitution logic — which gave India UPI instead of PayPal, and ONDC instead of platform monopolies — must be extended to AI. Indian models for agriculture, health, education and governance are both strategically important and commercially viable. Fourth, India's DPI partnerships with countries across Africa, Latin America, and Southeast Asia lay the foundation of an alternative digital order — one in which the Global South has a collective voice in setting the rules.

India's DPI story is genuinely world-class. But the journey is not complete. Building sovereign application-layer infrastructure while remaining structurally dependent on foreign cloud, foreign AI, and foreign chips is an unresolved vulnerability — one that the Nayara episode previewed and future geopolitical turbulence will test again. The countries that exercise genuine digital sovereignty in 2035 will be those that understand, in time, that sovereignty lives in the infrastructure — and not just in the apps built on top of it.

The writer is the former chairman of TRAI, secretary to the Government of India and CEO of the National Health Authority. Views are personal

5 Ebola patients recover as Congo opens new treatment centre

Associated Press
Bunia, May 31

FIVE PATIENTS have recovered from a rare type of Ebola virus, the head of the World Health Organization said Sunday during a visit to Bunia in eastern Congo, a city at the heart of an outbreak.

"Four people will be discharged today and there was one that was discharged the day before yesterday," WHO Director-General Tedros Adhanom Ghebreyesus said during the opening of a new Ebola treatment centre in Bunia, the capital of Ituri province.

"Of course, we're still working on vaccines and treatments but that doesn't mean that people cannot recover from Ebola," he added.

The WHO said Friday a patient had recovered from the Bundibugyo virus, the current species of Ebola, which has no approved treatment or vaccine. It was the first documented recovery of a confirmed Bundibugyo patient during the current outbreak. The health organisation said authorities have reported 134 confirmed cases in Congo and neighbouring Uganda, including 18 confirmed deaths as of May 29.

The Evangelical Medical Center in Ituri province, D R Congo on Sunday. REUTERS

Baraka Bulambulu, one of those who recovered, told The Associated Press on Sunday that community members feared contracting an unknown illness from them, keeping their distance while delivering food and medicine.

He said the uncertainty was overwhelming, as he and other patients believed they might die without knowing what disease they had, though testing eventually confirmed Ebola.

"Being able to come out of this alive is an immense source of happiness," Bulambulu said.

"Many people who were in the same situation died." Ezo Etienne, a nurse, said his symptoms began during ward rounds when he suddenly felt dizzy, then rapidly deteriorated into vomiting, intense itching, severe diarrhea and extreme weakness. He was tested seven times before Ebola was confirmed.

His treatment remained purely to treat the symptoms: medications to control vomiting, fluids to prevent dehydration and pain relievers. "That was all they could provide," he said. He urged the public and healthcare workers not to dismiss early symptoms, warning that misinformation leads many people to believe they have been poisoned rather than seeking hospital care.

● TOURISM IS A VITAL ECONOMIC PILLAR ACROSS ASIA, DRIVING JOBS, GDP GROWTH AND FOREIGN EXCHANGE EARNINGS IN COUNTRIES SUCH AS THAILAND, VIETNAM, CAMBODIA, THE PHILIPPINES AND NEPAL

Soaring prices during war jeopardize travel to tourism-dependent countries in Asia

Anton L Delgado & Chan Ho-Him
Bangkok, May 31

WITH SUMMER around the corner, soaring prices and other complications from the war with Iran are straining the tourism-dependent economies of countries in Southeast Asia, including Thailand and Vietnam.

The region's peak tourist summer season is at risk as elevated jet fuel costs coupled with ceasefire uncertainties prompt flight cancellations and higher ticket prices.

Tourism in Asia has yet to fully recover from the COVID-19 pandemic. Now, many countries are coping with the war's repercussions for global energy supplies and prices, which hit Asia first and hardest. Some families are pulling back on travel as visiting gas stations and grocery stores gets more expensive worldwide. Crowds have thinned at some places once synonymous with travel.

"With gasoline prices rising and tourism declining, how can we make money?" asked Siv Pech, a 58-year-old tuk-tuk driver in Siem Reap, home to Cambodia's centuries-old Angkor Wat temple complex.

Tourism is an economic lifeline for many developing nations. It contributes nearly 13% of gross domestic product in Thailand and nearly 9% in Vietnam, and it underpins millions of jobs in Cambodia. Travelers bring in much-needed foreign currency for import-dependent economies such as the Philippines and Nepal.

Those tourism dollars are more critical than ever as war-driven spikes in oil prices push up the cost of fuel imports, especially for parts of the world that relied on the Strait of Hormuz off Iran's coast as a conduit for much of their oil and gas.

The war will determine which tourism businesses can survive long enough to benefit from the eventual return of travelers, said Jitsai Santaputra of The Lantau Group, an energy industry consulting firm.

A crowd of tourists take pictures of a passing train at the Mae Klong Railway Market near Bangkok, Thailand on December 6, 2025. FILE

"This, happening within five years of each other, first the pandemic and now the war, is horrible for the tourism industry," she said.

Jet fuel shortages and surging costs have led Vietnam Airlines, the Malaysia-based AirAsia group, Hong Kong's Cathay Pacific and other carriers to cut flights or re-adjust schedules.

European carriers face a squeeze from similar issues.

Airspace closures across the Persian Gulf early in the war and the intermittent closures of certain Gulf airports cut off key layover locations for Asia-bound flights or forced commercial airplanes to take longer, costlier routes.

Airfares have jumped, with airlines like Air India and Cathay Pacific implementing sharp increases in fuel surcharges.

Cathay Pacific's fuel surcharge for medium-haul flights has jumped to 633 Hong Kong dollars ($80) from 264 Hong Kong dollars ($34) before the war. For long-haul flights, it increased to 1,362 Hong Kong dollars ($174) from 569 Hong Kong dollars ($73).

"Jet fuel prices remain at highly elevated levels" and have increased cost pressures, said Lavinia Lau, Cathay's chief customer and commercial officer. Travellers are booking closer to their departure dates, she said, indicating growing unease.

Sandra Awodele, a freelance travel writer in the Washington area, often plans year-round international trips and hoped this summer would finally be the one she crossed Asia off her bucket list.

In March, she began planning a long-awaited vacation to Thailand, envisioning one to two weeks of exploring. Her plans hit a wall when she checked airfares. "I looked at flight options and that's where it ended," Awodele said.

On the ground, rising fuel costs in tourism-dependent Southeast Asia are squeezing taxi and ride-hailing app drivers.

Pech, the Cambodian tuk-tuk driver, said he used to earn up to $20 a day toting tourists around Siem Reap. That's plummeted to about $5 a day.

His gas bill eats half of that. The rest goes to food. "Some days, I don't earn even a cent," he said. **AP**

• ENVIRONMENT

Aravalli range's role in shielding Gangetic plains from Thar dust

Sophiya Mathew
New Delhi, May 31

VISUALS OF a massive dust storm in Churu on Saturday have been going viral on social media. The storm, which affected Churu, Hanumangarh, Sri Ganganagar, Bikaner, Nagaur, Didwana-Kuchaman, Alwar and Sikar, is common in parts of Rajasthan this time of the year, when dust-bearing winds blow in from the Middle East and the Thar Desert. What stands between the Thar's dust and the densely populated Indo-Gangetic plains is the Aravalli range. As the winds run into the Aravallis, they lose speed and drop their load of sand. Many such sand dunes, called 'obstacle dunes', can be found on the range's slopes, visual proof of the protective role it plays.

As Chetan Agarwal, environmental researcher and Senior Fellow at the Centre for Ecology Development and Research, explained, "On many hills on the western slope, directly exposed to the hot winds from the west, there are large deposits of sand known as obstacle dunes. The vegetation growing in these dunes is similar to that found in desert regions. If there is additional tree cover, the wind has to pass through it, creating a natural scrubbing effect and reducing the movement of sand and dust. The dunes clearly show the role the Aravalli range plays in interception of dust."

This shield, however, is under stress, as the range degrades due to loss of vegetation and mining on hillocks. Meteorologists say dust from Rajasthan reaches the plains now even when a dust storm is not very intense.

Dust storms and the plains

Dust storms primarily originating from the Thar Desert are not unusual over northern India in the pre-monsoon months of April to June. These storms are fuelled by intense heat and dry conditions, while south-westerly and westerly winds move the dust across the region.

The India Meteorological Department's Climate Hazards and Vulnerability Atlas of India, based on 1981-2010 climatological normals, shows that parts of northwest India fall in the highest dust-storm frequency class, averaging 0.89 to 1.55 dust-storm days. The long-period data shows that Delhi and its neighbouring districts already lie in a high-exposure zone. With the Aravalli's degradation, the effects of such storms can be felt more.

"Nowadays, whenever wind speed reaches 35 to 40 kmph, dust from Rajasthan is carried up to the northern plains, which was not the case earlier. Usually, dust particles were carried when an intense dust storm originated over Rajasthan and travelled towards Delhi," said Mahesh Palawat, vice president of Meteorology & Climate Change at Skymet Weather Services Pvt. Ltd.

> **• WIDE-RANGING**
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> • Regular dust-carrying winds can alter the amount of sunlight reaching the ground, affecting surface temperatures.
>
> • NASA scientist Hiren Jethva said the effects of Aravalli's degradation and dust transport require long-term investigation and analysis.

State of the Aravalli range

A Forest Survey of India assessment from 2018 found that 31 of the 128 Aravalli hills in Rajasthan had disappeared due to anthropogenic pressures. There has also been a loss of hills with 200 metres to 600 metres elevation above sea level in Naraina, Kalwar, Kotputli, Jhalana, and Sariska in Rajasthan.

The Environment Ministry in its Aravalli restoration framework last year flagged the causes behind the degradation to be a mix of mining of red silica, granite and other minor minerals, deforestation, urbanisation, construction activities, land-use change, pastoral activities and encroachments.

A 2009 study by the Wildlife Institute of India, which researchers and officials still heavily cite, identified 12 gaps in the Aravalli range that have expanded due to forest degradation and inadequate cover.

These growing openings not only threaten to increase the frequency of sand and dust storms in surrounding regions including Delhi, Punjab, and Uttar Pradesh — and further deteriorate Delhi's air quality — but also impact rainfall.

• ECONOMICS

How China's reduced oil imports have helped India

Beijing's current reluctance to import oil is partly because high fuel prices and tepid economic activity have dampened demand

SUKALP SHARMA

THE WEST Asia war and the consequent closure of the Strait of Hormuz have constricted crude oil flows and sent prices spiralling, particularly for Asian economies such as India. The situation, however, could have been much worse had it not been for one country — China.

Beijing is the world's largest crude oil importer, but has sharply reduced its imports amid the crisis. This has freed up supplies from regions unaffected by the strait's closure — Russia, West Africa, the Atlantic Basin, and non-Hormuz-dependent West Asian nations — and allowed the likes of India, South Korea, Japan, Thailand, and Singapore to snap them up. So far, this has blunted the oil shock and prevented a deeper supply crunch and even higher prices in the region.

Why China cut oil imports

Before the war, 40-50% of India's oil imports came via the strait. India is the world's third-largest consumer of crude oil with an import dependency level of more than 88%.

"Chinese crude imports in May month-to-date are tracking around 6.6 mbd (million barrels a day), the lowest level since 2016. Compared with last year's average, imports from Russia, Africa, and parts of the Americas have declined materially, effectively freeing additional barrels into the broader Asian market," Sumit Ritolia, manager, modelling and refining at commodity market analytics firm Kpler, said in a recent note.

China's refinery crude intake has dropped to around 13.5 mbd in May — down 1.92 mbd versus the 2025 average — but the decline in oil demand continues to trail the drop in supplies. This means that China has been drawing on its huge onshore oil inventories. Preliminary data corroborates that trend.

Apart from its massive oil inventory and strategic crude petroleum reserves, Beijing's current reluctance to import oil at normal levels is partly because high fuel prices and tepid economic activity have dampened oil demand, while accelerating the energy transition in the country.

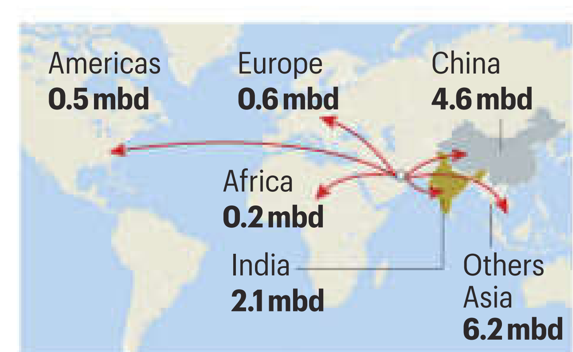
"...refiners across South Korea, Japan, Southeast Asia, and India have been able to secure more crude supply than initially expected, allowing refinery throughput to hold up materially better than earlier base-case assumptions despite tighter Middle Eastern flows and Strait of Hormuz disruptions. This has helped Asia (ex-China) refinery runs recover more strongly than initially expected in May, although overall refining activity remains heavily disrupted versus normal conditions," Ritolia said.

How China's reduced imports impacted India

India's crude imports in May have been strong. While the 2025 average stood at around 4.8 mbd, May imports are currently tracking around 5 mbd, potentially the highest monthly import in May ever, supported primarily by stronger Russian and Venezuelan crude imports, as per Kpler.

• Most crude via Strait of Hormuz goes to Asia

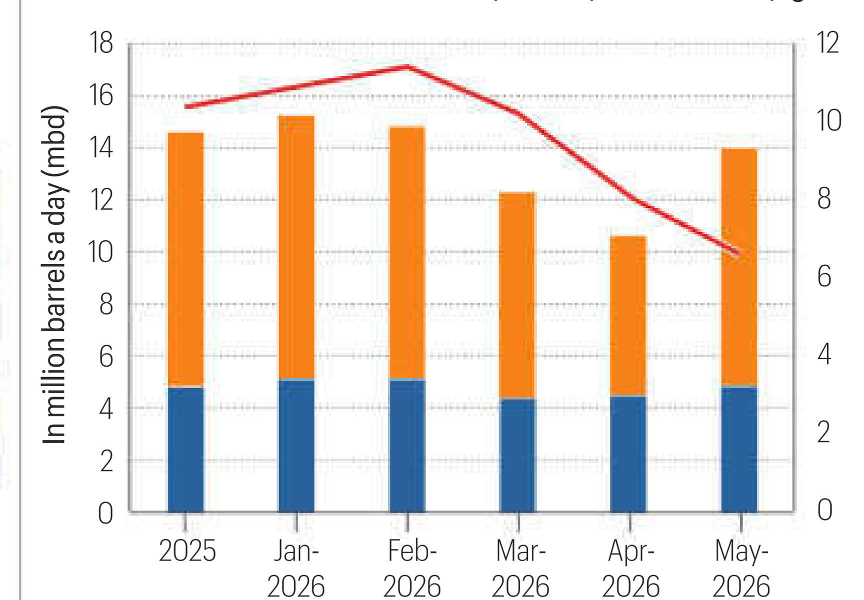
In 2025, nearly 15 million barrels a day (mbd) of crude oil, 34% of global crude trade, passed through the Strait of Hormuz. China and India combined received 44% of these exports.



NOTE: TOTAL FIGURE MENTIONED IN CHART INTRO DOES NOT MATCH THE SUM OF ADDING INDIVIDUAL NUMBERS DUE TO DESTINATIONS NOT INDICATED.
SOURCE: INTERNATIONAL ENERGY AGENCY ANALYSIS BASED ON KPLER.

• Sharp dip in China's crude imports

Chinese crude imports in May month-to-date are tracking around 6.6 mbd, the lowest level since 2016



NOTE: DATA AS OF 25TH MAY, SOURCE: KPLER

> **Impact on India**
>
> India is among the three Asian economies — along with Thailand and South Korea — most vulnerable to high oil prices, according to a Nomura report
>
> Every 10% oil price increase typically widens India's current account deficit by 0.4% of the GDP

Industry experts and analysts say that oil prices would have been a lot higher had China maintained its crude import levels.

"Physical crude oil benchmarks appear to have settled into a \$100-\$120/bbl (barrel) range over the past two months, after the initial wave of panic buying during the first month of the US-Iran war briefly pushed prices above \$150/bbl. Beyond SPR (strategic petroleum reserve) releases by multiple countries — particularly the accelerated drawdown in US stockpiles — one key reason oil prices have not sustainably moved above \$150/bbl despite a structural supply loss of 8 mbd from the Middle East is the absence of Chinese buying in the market," said Muyu Xu, senior crude oil analyst at Kpler.

Given its high oil consumption and import dependency, India is already bearing the brunt of high oil prices. The country imports 1.8-2 billion barrels of oil a year, and every \$1-per-barrel increase in price bumps up its oil import bill by up to \$2 billion on an annualised basis.

According to a March report by Nomura, India is among the three Asian economies most vulnerable to high oil prices, the other two being Thailand and South Korea. It also said that every 10% oil price increase typically widens India's current account deficit by 0.4% of the GDP. Crude oil alone is India's largest merchandise import.

According to Commerce Ministry data, crude oil imports in 2025-26 stood at about \$135 billion. If oil prices sustain at \$100 per barrel in the current financial year and import volumes don't decline, the oil import bill could be upwards of \$200 billion for the year. If the price sustains at a much higher level, the implications would be even graver.

Will China continue to hold back?

If Beijing's imports remain subdued, other Asian refiners are expected to continue benefitting from relatively improved oil availability in the foreseeable future. But any significant recovery could rapidly tighten oil supply in Asia — unless the Strait of Hormuz disruption is resolved.

The current pattern is expected to persist in the near term, as refiners continue to keep run rates low while staying on the sidelines of the physical crude market. According to Xu, Chinese state-owned refiners are heard to be planning to keep June and July refinery run rates largely unchanged, even as fuel demand typically starts to pick up during the summer season.

China's independent refiners have also cut back significantly on buying oil from the spot market amid a surge in Russian and Iranian oil prices, which formed a large part of these refiners' crude slate.

"That said, if the US were to maintain the blockade on Iranian cargoes, Chinese independent refiners would have to turn to alternative supplies — most likely Russian crude — and compete for barrels currently taken by other buyers such as India. Given elevated prices, they are more likely to further cut run rates," Xu said.

So, there could be further oil inventory drawdowns in China, which may prompt Beijing to release state oil reserves, or even step up crude purchases. Unless oil flows through the strait normalise, China will have to take a call on when to step up oil imports, which is when the real oil shock would be visible.

• FINANCE

Unaffordable homes: UN report lists factors behind housing crisis

Devansh Mittal
New Delhi, May 31

ACROSS COUNTRIES, housing prices are slipping out of the budget of large sections of the population. The world is in the midst of a housing crisis as more people move to urban areas and not enough houses are being built, according to a recent report by the United Nations. The report also flags financialisation as one of the major factors behind this crisis.

What is the global housing shortage?

Up to 3.4 billion people — four out of every 10 people worldwide — lack access to adequate housing. Of this, more than a billion live in informal settlements such as slums, according to the report by the United Nations Human Settlements Programme (UN-Habitat), the highest recorded number.

The report was released at the 13th World Urban Forum, organised by UN-Habitat in Baku, Azerbaijan.

If a household spends more than 30% of its budget on housing, it is considered stress burdened. Globally, 44% of households spend more than 30% of their income on rent.

For prospective buyers, affordability is measured using the house price-to-income ratio: housing is considered affordable when the median house price is no more than three times the median household income, while ratios above five indicate severe unaffordability.

Central and Southern Asia, including India, has seen the sharpest relative rise in price-to-income ratios — up 73% between 2010 and 2023. In 2018, more than half of new housing units built in the eight largest Indian cities were in the affordable housing segment. This has declined to fewer than two out of every ten units built by 2025, the report notes. Developers prioritise luxury units due to higher margins and demand, placing homeownership out of reach of most families.

What are the roots of the crisis?

The roots of the crisis go back to the post-World War II period, the report notes. Facing severe housing shortages and economic depression, governments invested heavily in public housing. But international agencies, including the United Nations, the International Labour Organisation (ILO) and World Bank, rarely supported this model. Instead, they advocated greater private sector involvement, arguing that mass public housing was too costly, especially for newly independent and transitioning economies.

This critique of public housing laid the foundations of two dominant approaches that continue to shape housing policy: "aided self-help" — where households build or improve homes with support — and the "enabling approach", under which governments facilitate construction by private players, community associations and others. Yet these programmes often failed to reach the poorest households and sometimes displaced the very populations they intended to support, the report states.

> **Multiple reasons**
>
> • In some cases, well-intentioned housing subsidies also contributed to price increases by raising purchasing power without expanding supply.
>
> • Declining interest rates from the 1980s further encouraged investment in real estate, intensifying demand and price growth.

The state has not only built fewer homes but often built them in the wrong locations. An analysis of annual Delhi Development Authority (DDA) reports by the Centre for Policy Research's Cities of Delhi project, directed by Partha Mukhopadhyay and Patrick Heller, found that the agency consistently fell behind housing targets. "We calculate that of the 979,073 houses built between 2003 and 2010, fewer than 23,000 (2.3%) were built by the DDA," it said.

The report also notes that housing must be developed where livelihoods and social infrastructure such as parks exist, otherwise people are unwilling to relocate. More than 50,000 government-built flats remain unoccupied on Delhi's outskirts due to poor connectivity, lack of livelihoods and political tussles.

What is financialisation and what are its impacts?

According to the report, financialisation is one of the major drivers of the crisis. It refers to the increasing involvement of financial actors, investment instruments and large pools of capital in housing and land markets. In India, Non-Resident Indians (NRIs) and investors purchase luxury housing as investment.

Access to home loans in India increased after liberalisation in 1991 made homeownership more widely accessible.

However, the report states that globally this increased availability of credit allowed lending to grow faster than the construction of new homes. This created loops in which easier access to credit pushed up prices more quickly than supply could respond.

The effects of financialisation vary depending on local conditions, regulation and supply. In markets with limited supply, informal incomes or weak tenant protections, increased capital flows can lead to evictions, rent hikes and speculative redevelopment.

The report also notes that in better-regulated environments with stronger planning systems, investment has supported new construction, upgrades of older housing, and professionally managed rental stock.

The reporter attended the World Urban Forum in Baku on the invitation of UN-Habitat.

• POLICY

Religion and the tribal identity: Why 'delisting' debate refuses to die down

Shubham Tigga
Ranchi, May 31

WHEN THE Census process started in Jharkhand on May 16, an appeal began circulating among Adivasi communities: don't mention "Hindu" in the religion column; instead, write "Sarna", the traditional tribal faith, or "others". Around a week later, a large Adivasi gathering in New Delhi demanded "delisting" from the Scheduled Tribes (ST) category for tribal people who convert to Christianity or Islam.

These two recent developments have revived the long-running debate around religious identity among Adivasis, not only in Jharkhand but elsewhere too.

The ST status

Scheduled Tribes in India are those recognised under Article 342 of the Constitution as eligible for certain benefits and State support. "Delisting" refers to the demand that tribal people converting to Christianity or Islam should not continue receiving the benefits for STs. Among the major voices raising this demand are Adivasi Hindus.

But another section of Adivasis, from the Sarna community (which does not identify as either Hindu or Christian), argues that if religion becomes the basis for removing tribal status, the same rule should apply to tribal people who have converted to Hinduism.

This comes amid the recent Supreme Court observation reiterating that Dalits converting to Christianity or Islam cannot continue to claim Scheduled Caste status under Article 341. However, Article 342, which deals with STs, does not explicitly mention religion.

The debate

The Janjati Suraksha Manch (JSM), affiliated with the Rashtriya Swayamsevak Sangh's (RSS) tribal outfit Akhil Bharatiya Vanvasi Kalyan Ashram, organised the New Delhi gathering on May 24. Attended by Union Home Minister Amit Shah, it saw several Adivasi groups, including some from parts of the Northeast, backing the delisting demand.

However, the Sarna community and the Christian Adivasi Mahasabha organised a similarly large counter-rally in Chhattisgarh's Jashpur district. Adivasi activists from Jharkhand and Chhattisgarh, especially followers of the Sarna faith, extended support to Christian Adivasis and distanced themselves from organisations seen as close to the RSS.

The Delhi event, notably, triggered another controversy. Home Minister Shah used the word "*vanvasi*" while talking about delisting, leading to protests in Jharkhand. Protesters argued that the word, meaning "forest dweller", reduces tribal identity only to geography. "Adivasi", which means "original inhabitants", carries a political and historical identity linked to land and indigeneity.

Origins of the demand

The roots of the delisting idea are often traced to Baba Kartik Oraon, a prominent Adivasi leader who later became a Union Minister in Indira Gandhi's Cabinet.

The event in Chhattisgarh's Jashpur district on May 24. EXPRESS

Kartik Oraon lost his debut Lok Sabha election in 1962 from Lohardaga, an ST-reserved constituency. He argued before the Patna High Court that since his opponent had converted to Christianity, he no longer was a tribal from the Oraon community and should, therefore, be disqualified.

The Patna High Court, however, held that Oraon is "primarily a tribe and ethnic identity, not merely a religion". It observed that Christian Oraons retained their clan system, and continued to observe tribal customs and festivals. The judgment noted they are "Oraons first and Christians next".

This High Court judgment is often cited by opponents of the delisting demand to argue that religion is immaterial in determining ST identity.

In August 1967, when Kartik Oraon was a minister in Indira Gandhi's Cabinet, the government introduced the 'Scheduled Castes and Scheduled Tribes Order Amendment Bill' in Lok Sabha to revise the inclusion and exclusion of castes and tribes across India. It was referred to a Joint Parliamentary Committee the next year in March.

The committee, in its report on November 17, 1969, proposed amending the Bill to exclude Christian and Muslim tribal converts from the Scheduled Tribe category. The government, however, expressed reservations over the proposal. Parliament never adopted it.

Tribal people and groups affiliated to the RSS and BJP have often invoked Kartik Oraon to claim that Adivasis are Hindus and cannot remain Christian while enjoying Constitutional protections for STs.

Kartik Oraon himself, however, had a different view. In his book, *Bees Varsh Ki Kaali Raat*, he defines Adivasi faith as "Adi Dharma", an indigenous religion that existed before Hinduism. While he does argue that converted Christians should not continue receiving ST benefits, he states that Adivasis are not Hindus, questioning under which *varna* (Brahmin, Kshatriya, Vaishya or Shudra) they could be placed within the Hindu social order.

LONGER VERSION
INDIANEXPRESS.COM/EXPLAINED

ANALYSTS EXPECT MPC TO LOWER GROWTH PROJECTIONS FROM ITS EARLIER FORECAST OF 6.9%

Reserve Bank may defer rate hike despite inflationary pressures

George Mathew
Mumbai, May 31

THE RESERVE Bank of India's (RBI) Monetary Policy Committee (MPC) is widely expected to remain in a wait-and-watch mode on June 5 and defer any move to hike interest rates despite mounting inflationary pressures from rising crude oil prices.

RBI policymakers face a complex balancing act as the ongoing conflict in West Asia threatens to push up energy costs, disrupt global supply chains, and trigger volatility in financial markets, at a time when sustained capital outflows and pressure on the rupee add to macroeconomic concerns.

Against this backdrop, the RBI is likely to reassess both its growth and inflation projections, with risks emanating from elevated oil prices, uncertainty in global trade, adverse weather conditions and geopolitical tensions clouding the economic outlook.

No tightening now, but later this year?

The RBI had kept the main policy instrument, the repo rate (interest rate at which it lends money to commercial banks), unchanged at 5.25% in the April policy review. In doing so, it underscored a cautious and calculated approach, as economic risks resulting from the West Asia conflict have darkened both global and domestic outlooks.

In the absence of a clear risk of headline inflation spilling over into core inflation (a measure that excludes the change in prices of food and fuel) and signs of unanchored inflationary expectations, the central bank is likely to view the energy shock as a supply-side price catalyst. It could defer tightening policy at the upcoming meeting, as per Radhika Rao, Senior Economist and ED, DBS Bank.

"An argument can, nonetheless, be made on the need to tighten policy in the second half of calendar year 2026 if the conflict continues, to attract rate-sensitive flows," Rao said. "Add to this, the inflation rationale has also gained credibility amid successive pump price increases (7% over the past two weeks), a pickup in food, impact of prevailing heatwave conditions and rising business inflation expectations, all of which point to mounting underlying price pressures," Rao said.

CareEdge Rating said it expects the MPC to adopt a wait-and-watch approach amid external volatility, assessing whether current price pressures are transient or persistent, based on external developments. India's still-benign retail inflation, at 3.48% in April and below the RBI's 4% medium-term target for more than a year, gives the central bank room to stay on hold rather than rush into policy action.

• INFLATION DYNAMICS

AN ARGUMENT can be made on the need to tighten policy in the second half of calendar year 2026 if the conflict continues, to attract rate-sensitive flows, says Rao

INFLATION RATIONALE has gained credibility amid successive pump price increases (7% over the past two weeks), pickup in food, impact of prevailing heatwave conditions and rising business inflation expectations, all of which point to mounting underlying price pressures, she said

FUTURE TRAJECTORY of policy rate will depend on MPC's assessment of evolving inflation dynamics, which are significantly influenced by external factors

Growth may be slashed

Economic growth in FY27 is expected to remain below the economy's estimated potential growth rate of 7%, as projected by the Economic Survey. This strengthens the case for the RBI to remain comfortable with a lower real policy rate for the next few quarters, particularly as growth continues to require policy support.

Unless there is clear and sustained evidence that higher input costs are feeding into household inflation expectations, the central bank is unlikely to go for a rate hike. The future trajectory of the policy rate will depend on the MPC's assessment of evolving inflation dynamics, which are significantly influenced by external factors. If conflict persists and inflation risks become entrenched in household expectations, rate hikes could be considered toward the end of the year, CareEdge Rating said.

Analysts also expect the MPC to lower growth projections from its earlier projection of 6.9%. "MPC will also remain attentive to growth risks. We project FY27 GDP growth at 6.7%, assuming crude oil averaging $90. However, prolonged conflict and oil prices around $110 could lower growth closer to 6%," CareEdge said. It expects Q4 FY26 GDP growth to moderate to around 6.8-7.0%, compared with 7.8% in Q3FY26.

Q4FY26 remained partially insulated from the adverse effects of the crisis, as the first two months of the quarter were relatively unaffected.

The RBI's Annual Report said the West Asia conflict and risks such as higher energy prices, supply chain disruptions, financial market volatility, global trade uncertainty and weather disruptions could affect growth and inflation in the short term. In a highly uncertain global environment, continuous assessment of the evolving developments is warranted to frame the appropriate policy response on an ongoing basis, the central bank said.

FULL REPORT ON WWW.INDIANEXPRESS.COM

Rupee defence to liquidity: RBI earned over Rs 4 lakh cr in FY26, spent a third of it

Siddharth Upasani
New Delhi, May 31

LAST WEEK, the Reserve Bank of India (RBI) transferred a record dividend of Rs 2.87 lakh crore to the Centre for 2025-26. Unlike companies that transfer a portion of their profits to shareholders, the RBI must hand over its entire gain to the government. But how did it end up generating such a big profit?

First and foremost, the RBI's objective is not to make a profit; it just so happens that the actions it must take to achieve its objectives — ensuring adequate supply of money (liquidity) in the banking system, appropriately investing its foreign exchange reserves, and maintaining order in the bond and foreign exchange market — result in a lot of revenue.

In 2025-26, the RBI's total income was a record Rs 4.28 lakh crore, up 26% from the previous year. Of this, it spent about a third, or Rs 1.41 lakh crore. But how did the RBI earn so much and what did it spend on?

Supporting the rupee

Last year was highly turbulent: tariffs, wars, and the AI boom led to foreign investors dumping Indian shares and debt, while India's gold and silver import bill hit new highs. This led to the rupee falling by 10% against the US dollar. The byproduct of the RBI's attempts to defend the rupee was the largest income stream for the central bank.

The RBI shores up the rupee by selling previously-bought foreign currency. The difference in cost and selling price represents a pure profit. For instance, if a dollar was bought for 75 rupees and sold at 95, that's a profit of Rs 20. All in all, the RBI sold $195 billion of foreign currency on a gross basis in 2025-26.

This led to what is called an 'exchange gain from foreign exchange transactions' of Rs 1.69 lakh crore, 52% higher from 2024-25.

The RBI does not disclose the average historical cost of the foreign currency it holds or sells.

• How RBI earned and spent money in FY26

	2024-25 (Rs cr)	2025-26 (Rs cr)
Interest from rupee securities	85,525	117,740
Interest from foreign securities	97,007	107,679
Interest on foreign deposits	36,195	27,407
Net interest on liquidity operations	-10,120	-19,163
Exchange gain from forex transactions	111,143	168,906
TOTAL INCOME	**338,308**	**427,684**
Printing of notes	6,373	4,875
Employee cost	9,147	10,136
Provisions	44,862	109,380
TOTAL EXPENDITURE	**69,714**	**141,092**
Dividend to government	**268,590**	**286,588**

NOTE: INTEREST ON LIQUIDITY OPERATIONS INCLUDES LAF, SDF, MSF SOURCE: RBI

Consequences of rupee defence

Selling forex has consequences which the RBI must negate. And these actions generate more income.

When the RBI sells dollars, it gets rupees in return. This reduces the availability of Indian rupees — or, rupee liquidity. Unaddressed, this rupee liquidity shortage would raise market interest rates.

In 2025-26, the RBI did not want interest rates to rise (it cut the repo rate by 125 basis points). So, it had to infuse rupees into the Indian market to nullify the impact of its dollar sales. It did so by buying central government bonds worth nearly Rs 9 lakh crore from those who hold them – banks, for instance. These bonds, and those it already held, provided an interest of Rs 1.18 lakh crore to the RBI last year.

While these government bond purchases infuse long-term liquidity, short-term mismatches require the RBI to lend and hold banks' money. On the whole, these short-term operations saw the central bank pay out Rs 19,163 crore as interest.

The RBI also earns interest on the foreign securities it holds as part of its foreign exchange reserves. In 2025-26, this interest income rose 11% to Rs 1.08 lakh crore. It also received Rs 27,407 crore as interest on its foreign deposits.

Cost of operations

Two expenditure items are straightforward but minor in the overall scheme of things. Printing of new currency set back the RBI Rs 4,875 crore, while employee cost rose 11% to Rs 10,136 crore.

The big chunky expenditure was provisions. Like any other bank, the RBI must set aside money if its loans — in the RBI's case, investments — are doing poorly. This happens when the value of its holding of domestic and foreign securities falls due to a rise in yields, which is what happened in 2025-26.

Money also had to be set aside for the losses being incurred by the RBI on its forward contracts — another instrument used to defend the rupee last year. All put together, the net provision made by the RBI on account of these three issues was a massive Rs 1.09 lakh crore.

The provision would have been higher had the Contingent Risk Buffer not been cut to 6.5% of the balance sheet from 7.5%.

India, Oman FTA to come into effect today

Mukesh Jagota
New Delhi, May 31

THE INDIA-OMAN Comprehensive Economic Partnership Agreement (CEPA) will take effect on Monday, extending zero-duty access on 98% of tariff lines, which cover 99.4% of India's exports to Oman. In return, India will open up 77% of its tariff lines for Oman, covering 94% of its total exports to India. This is India's 15th bilateral trade agreement. For India, the most visible gains lie in merchandise exports. Oman's gains are concentrated in energy and industrial inputs, said Ajay Srivastava, founder of GTRI. With over 6,000 India–Oman joint ventures and Indian investments exceeding $7.5 billion — mainly in the Sohar and Salalah free zones — the CEPA is as much a strategic pact as a trade deal, he added. **FE**

• PREVAILING GLOBAL UNCERTAINTY IS DRIVING ADOPTION OF MULTI-CURRENCY FRAMEWORKS, WITH THE EURO AND SWISS FRANC EMERGING AS PREFERRED ALTERNATIVES, SAYS UBS

Ditching the US dollar: How family offices are reacting to geopolitical risks

George Mathew
Mumbai, May 31

THE SWISS bank UBS, a global wealth manager, stated in a report on Thursday (May 28) that many family offices are considering reducing their exposure to the US dollar or diversifying regionally amid rising geopolitical risks.

A family office is essentially a private wealth management advisory firm that serves ultra-high-net-worth individuals (UHNIs) and their families in managing and growing their wealth across generations.

Often, family members hold a share in most of the family offices' assets. In India, too, outbound investments routed through family offices for HNIs are now a topic of discussion amid the broader outbound fund flows.

Drawing on insights from 307 family offices across more than 30 markets, the report highlighted a notable shift in currency positioning. "65% of family offices expect confidence in the US dollar's reserve status to weaken, with many reassessing exposure to USD-denominated assets," the report said.

Another major theme was artificial intelligence (AI) as the leading global investment.

UBS said 65% of family offices have already invested across the AI value chain, including data centre infrastructure, software platforms and producers of semiconductors.

65% of family offices expect confidence in US dollar's reserve status to weaken. REUTERS

• GLOBAL SHIFT

- Drawing on insights from 307 family offices across more than 30 markets, the report highlighted a notable shift in currency positioning
- Another key theme was AI as the leading global investment. 65% of family offices have already invested across the AI value chain.
- North America accounts for the largest share of allocations among geographies, yet family offices seek to reduce concentration risk

Slight shift from dollar, North America still key

Each of the family offices analysed in the report managed an average of $1.3 billion in assets. Participating families had an average net worth of $2.7 billion, and their total wealth amounted to $627.4 billion, UBS said. It found that prevailing geopolitical uncertainty, especially with recent events in West Asia, is driving broader adoption of multi-currency frameworks, with the euro and Swiss franc emerging as preferred alternatives.

Among geographies, North America continues to account for the largest share of allocations, yet family offices are actively seeking to reduce concentration risk. Increasingly, they are planning to expand exposure to Asia Pacific, Greater China and Western Europe, reflecting a structural shift.

UBS found that family offices in the Middle East demonstrated the highest level of planned portfolio change globally, with 82% intending to adjust strategic allocations. Their portfolios remain anchored in North America (50%), but with meaningful exposure to Western Europe and the Middle East, reflecting a hybrid investment approach. Family offices in the US exhibit the strongest home bias globally, with 88% of portfolios allocated to North America. This implies confidence in the depth, liquidity and resilience of domestic capital markets.

Rise of AI, despite bubble concerns

While AI remains the top investment theme (65%), those surveyed also show heightened interest in defence and security infrastructure (39%) and broader infrastructure investments (35%). It may potentially reflect geopolitical considerations alongside growth opportunities, the report said. There are also concerns that certain AI stocks are overvalued at the moment, and an 'AI bubble' could burst when actual revenues fall behind the market hype. "AI continues to stand out as the defining investment theme of this decade," said Yves-Alain Sommerhalder, Head of Global Wealth Management Solutions at UBS. "Family offices are approaching it with both conviction and selectivity, seeking opportunities across the value chain while balancing long-term growth potential with risk discipline," he added.

Family offices are allocating funds towards power and resources (37%), infrastructure (37%), and AI-enabled healthcare (33%), recognising the broader ecosystem required to support and scale AI adoption, according to the report. By contrast, crypto and digital assets remain a niche allocation, with only 24% of family offices investing in them, typically at low single-digit levels. Allocations are usually modest (around 1%).

Gaps in governance, succession planning

While family offices continue to professionalise investment operations, the report highlights persistent gaps in governance frameworks, succession planning and next-generation engagement, creating potential risks to long-term continuity. Operationally, many family offices have adopted institutional-grade practices: 68% have formal financial performance measurement processes, 60% operate with investment committees, and over half use structured budgeting frameworks, reflecting a growing level of rigour and oversight, UBS said. However, this progress is uneven. Fewer than half have implemented formal governance frameworks with board-level oversight, and only 35% have a defined succession plan for the family office itself.

FULL REPORT ON WWW.INDIANEXPRESS.COM

NEWSMAKERS IN THE NEWSROOM

WHY MONTEK SINGH AHLUWALIA

Few understand India's political economy as Montek Singh Ahluwalia does. After his graduate studies at Oxford and a stint in the World Bank, he returned and served the government in various capacities; as Commerce Secretary and Finance Secretary in the early 1990s, he helped shape India's response to the balance of payment crisis of 1991, setting in motion its transformation into a market-linked economy by dismantling the licence raj and ushering in reforms across sectors. As Deputy Chairman of the Planning Commission in the UPA government (2004-14), he had the then Prime Minister Manmohan Singh's ear, and coordinated India's actions following the 2008 global financial crisis. He is sharp, and can sense what investors are looking for in the India story.

'FTAs with Europe, UK good moves... should ensure industry doesn't put pressure to take rigid positions'

Economist Montek Singh Ahluwalia on the need for a strategy for managing the balance of payments, price signals being suppressed, managing exports and why rupee depreciation, for the moment, is a good thing. The session was moderated by P Vaidyanathan Iyer, Managing Editor, *The Indian Express*

ON NEED FOR NEW BILATERAL INVESTMENT TREATIES

We have a reputation of being 'investor unfriendly' because of retrospective taxation and arbitrary changes of policy which are often adverse to foreign investors

ON COMPANIES INVESTING ABROAD

Setting up subsidiaries abroad is a natural way for them to expand. If in the next 20 years we don't see more Indian manufactured products abroad, then the Indian private sector can be said to have failed the country

P Vaidyanathan Iyer: Could you take us from 2000 onwards, the first 13-14 years and the next 13 years?

We have had three different governments in this period. The Vajpayee NDA from 1998 to 2004, the UPA from 2004 to 2014 and the current Modi NDA from 2014 to today. It is fair to say that all three have talked about the need for reforms and also made some progress, broadly following the direction set in 1991. However, the 1991 reforms were transformational, dismantling decades of license permit raj controls. What has happened since then is much more gradualist. The fact that governments under different political parties moved forward broadly in the same direction is good news. It shows that there is a broad consensus beneath political polemics. It is particularly important that parties opposing reforms when in Opposition, adopted them when in office, as happened in the case of Aadhaar and also with the GST. However, the pace of movement has been far too slow. I think we are now at a stage when this sort of gradualism will not be enough.

P Vaidyanathan Iyer: So, what's the India story now?

The world is preoccupied with too many things and India is not a focus of attention. This is in some ways an advantage. But we should try to reestablish a positive India story.

The positive features about India's economy are well-known. We have favourable demographics; the economy grew at 8 plus per cent over an extended period before 2010 highlighting the strengths of the Indian private sector; infrastructure and logistics are improving though more needs to be done; the digital public infrastructure has expanded over the past 10 years, leading to an increase in financial inclusion.

All this is good but it is not enough. To reestablish the India story we need a much clearer articulation of how we intend to deal with the immediate challenge we face as a result of the oil crisis and the longer run challenge of how to accelerate growth to achieve Viksit Bharat.

P Vaidyanathan Iyer: Let's start with the balance of payments. How do you see India's prospects given there's no immediate end to the war? Does the three years of continuous pressure on the balance of payments worry you?

As you say there is no immediate end to the war in sight but it does seem as if both the US and Iran want to avoid escalation. There is still considerable uncertainty about when oil supplies will return to normal but earlier fears of oil going beyond $120 have subsided. Even if we assume that the US and Iran reach a satisfactory agreement in the next two weeks, it will still take time for movement through the Strait of Hormuz to get back to normal. And since many countries have drastically reduced their strategic stocks, they will want to build them up as soon as possible.

All this means oil prices which, are currently below $100 per barrel which is a third higher than in the previous year, may remain at that level for some time. We need to outline a credible strategy for managing the balance of payments in this situation.

As you point out, we had a problem even before the oil price shock. For the past several years, we have run a relatively modest current account deficit averaging around 1.5 per cent of GDP, which was more than offset by a capital account surplus of over 2 per cent of GDP. That surplus declined sharply after 2023 and has almost disappeared in 2025-26. We have seen sharp fall in net Foreign Direct Investment (FDI) and the net flow of foreign portfolio investment has turned negative.

The disappearance of the surplus on capital account would have posed problems even without the rise in oil prices. As it is the higher oil prices will worsen the current account deficit could worsen to say 2.2 per cent of GDP in the 2025-26. To manage this with low capital flows, we have to cut the current account deficit and increase capital flows as much as possible. If things look as if they are going to come back to normal sooner than people think, we could allowing the reserves to take some of the hit.

A conventional response when trying to reduce the current account deficit is to allow the currency to depreciate. This has happened over the past year and I think the RBI did the right thing by allowing the depreciation to take place. This is not to say that continuous depreciation will solve the problem. We have to avoid creating self perpetuating expectations, which can worsen the situation. That is best done by taking other actions to contain the current account deficit and also taking steps to revive capital inflows as much as we can.

A second sensible response is to let the increase in oil prices be passed through into the economy. That too has happened. The increase is modest and the government has also taken some of the hit by lowering excise duties.

P Vaidyanathan Iyer: What else should we do?

Our performance in export of goods has been quite poor for the past several years and this weakness needs to be addressed.

Economist Montek Singh Ahluwalia (right) in conversation with P Vaidyanathan Iyer, Managing Editor, *The Indian Express*, at the Noida office ABHINAV SAHA

Vietnam and Bangladesh, benefited from the China Plus One opportunity, but we did not. The root problem is that Indian industry has been excessively shielded from international competition making us less competitive in exports. The problem was worsened because import duties were raised on a number of items from 2016 onwards. These were inputs of MSMEs which also do a lot of exports. Since the rupee has weakened, these duties should now be lowered. It will help MSMEs and exporters.

The FTAs signed with Europe and the UK are very good moves, but they will only become operational when many details that need to be resolved are agreed upon. We should ensure that domestic industry does not put pressure on our negotiators to take rigid positions on these details on protectionist grounds. We should obviously protect substantive national interests but we should ensure that the FTAs become operational as soon as possible.

The Asia region is projected to be the fastest growing region in the years ahead, and we need to integrate more with this region. We decided not to join RCEP in 2015 probably because Indian industry was unwilling to face competition from China. We should now apply to join the Comprehensive and Progressive Agreement for Teans Pacific Partnership (CPTPP) where China is not a member. It will send a clear signal that India is determined to integrate more fully with the world economy.

The fall in capital flows also calls for corrective action but what we can do in the short run is limited. Portfolio investment is known to be volatile and Indian P/E ratios were over valued earlier. We can only assume these flows will come back as the world economy normalises and the India story comes back into play. The fall in net FDI is more worrying and calls for some reevaluation of our policy. An area where we can take early steps relates to improving the ease of doing business.

Some months ago, newspapers reported that the Committee examining the ease of doing business under Rajiv Gauba (former Cabinet Secretary and now Member NITI Aayog) had recommended withdrawal of almost 700 Quality Control Orders (QCOs). I thought this was an excellent recommendation but I believe only about 200 QCOs have been removed. QCOs are a disguised form of the import license permit raj. It would send a very good signal if the government implemented the Gauba Committee recommendation in full. This can be done immediately. Open-ended reliance on QCOs creates uncertainty about policy.

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ON FUEL PRICE DEREGULATION

As the number of vehicles increases, we need to ensure that more of them are electric vehicles (EVs). Raising the price of petroleum products is a way of encouraging EVs. It is better than subsidising them

The unilateral decision in 2015 to scrap all existing Bilateral Investment Treaties (BITs) may also account for what has happened to FDI flows. Investments already made under the old BITs would continue to receive protection for some years, but for all new investments there is no protection. It was announced a year ago that a new BIT would be released but it has not yet happened.

The absence of a BIT may not matter when the country is experiencing high growth, FDI is flowing in, and investors have a good story to tell. It is a different matter if there are more exits than entries. The negative effect is magnified if we have acquired a reputation of being 'investor unfriendly' action, because of retrospective taxation, and arbitrary changes of policy which are often adverse to foreign investors but favour domestic competitors.

Foreign investors make a detailed assessment of the 'India risk' before investing in India and this includes the risk of getting into tax disputes. They have all heard of Mohandas Pai, a well-known investor who has complained of 'tax terrorism'. This refers to arbitrary interpretations and the likelihood that if you get into a tax dispute, the government will pursue it all the way to the Supreme Court, which will take 20 years. Fixing this problem will help not only FDI but also domestic investors.

I personally think there is merit in having these issues examined by an external high-level committee, headed by someone outside the government, which includes two or three business people, tax lawyers and consultants. We could also have retired secretaries from the Finance Ministry — retirement frees them from positions they may have taken earlier, which they cannot realistically ignore while in service. This committee could be charged with discussing with business groups and making specific recommendations in a report which is made public for wider discussion.

Whatever is accepted could then be incorporated in the next budget. I don't think this sort of revamping can be done by internal committees of officials, whose reports are generally not made public.

Siddharth Upasani: You talked about markets realising the exchange rate has been overdone. But aren't price signals across the economy somewhat suppressed, be it fuel prices or in agriculture?

This is a problem. We had announced dismantling of the administered price system in petroleum products in 2002 but it never really happened. Electricity prices are also subject to extensive administered controls determined by state regulatory commissions, which are heavily influenced by state governments. Fertiliser prices are grossly distorted at present with urea being sold to farmers at 10 per cent of the cost of imported fertiliser. We need to review these controls and come up with a policy where these prices respond to global market conditions. Where the resulting price would hit vulnerable sections, we should compensate them through a direct cash transfer. Fortunately, the Aadhaar-based digital system now makes it possible to do this. These changes can be made over a five-year period but continuing with gross distortions in key prices is not feasible.

Managing water scarcity is another problem since water is not priced. Some of what we are doing, for instance, expanding use of methanol to mix with petrol and diesel may not be economically viable if water is properly costed. Large expansion of data centres will also come up against constraints of available water. These are very important problems which need to be spelt out and addressed.

Aakash Joshi: What did you make of the piece Surjit Bhalla wrote for *The Indian Express* (on May 23)?

Surjit holds the bureaucracy — which he calls the 'deep state' — responsible for perpetuating policies which prevent investors from investing in India. I don't agree with his deep state description, but I agree that we need to review our policies to see why they discourage investors. That's why I have recommended setting up a high-level committee, drawn from outside the government with its report being released for discussion. Relying on inside groups, whose reports are not made public cannot yield results. Some of the problems which concern foreign investors also trouble domestic investors, so solving them will be a double whammy.

Anil Sasi: One reason for the adverse net FDI inflows is that our manufacturers are not investing here, but outside.

I don't think we should restrict our companies from investing abroad. Many Indian companies in the IT sector, pharmaceuticals and also others are working in a highly competitive global environment and they want to expand their footprint. Setting up subsidiaries and acquiring companies abroad are natural ways for them to expand and we should not stop it.

In the longer run, this will lead to a larger presence of Indian products and brand names abroad. That has not yet happened, but it is the next stage. If in the next 20 years we don't see any many more Indian manufactured products abroad, then the Indian private sector can be said to have failed the country. But we should not end up in a situation where they say 'we were keen to do this, but you didn't allow us to expand abroad'. Many people say that some of the outward investment may represent Indians wishing to leave the country. We should look into why they want to do this and correct the domestic policy failures that generate this behaviour.

Liz Mathew: How much do you think economic issues influence a voter?

That is a very good question. As an economist, I always hope that politicians can sell a policy which will yield results in the longer run even if is not a winner in getting votes. So politicians have to select a package which also provides some tangible results in the short run. I know it is not easy, but that is the real job of politicians.

Sukalp Sharma: Given the global energy shock, do you feel it would be prudent if the government implemented demand suppression measure on fuel consumption?

If by demand suppression you mean raising fuel prices, yes that is the way to go. We have raised prices a little and the government has also taken a hit. They may have to do more if oil prices move up. If however, you mean 'managing demand' through some form of rationing, I feel that would not be workable. You cannot do it by saying only 15 litres will be available at the petrol pump at any given time. It will only lead to use of multiple pumps. You would have to issue entitlement certificates against which petrol and diesel can be bought. That is an impossible administrative task and will also be a disaster. You can be sure that such permits if issued to users would be sold in the black market.

Sukalp Sharma: You spoke of how fuel price deregulation is only on paper. Is a deregulated system better in times of volatility?

As I said, we announced that the administered price system was dismantled in 2002 but we didn't really deregulate. I think as a matter of policy we should implement effective deregulation. Remember that as the number of vehicles increases, we need to ensure that more and more of them are EVs. Raising the price of petroleum products is a way of encouraging EVs. It is better than subsidising them. Perhaps, the NITI Aayog should prepare a discussion paper outlining how we can move to deregulated petroleum prices over say, a period of five years.

THE INDIAN EXPRESS

Your Health

MONDAY, JUNE 1, 2026

{ WEEKLY CAPSULE }

Drug to stop fatty liver disease

SCIENTISTS AT the University of San Diego have unveiled a potentially game-changing treatment for people with metabolic dysfunction-associated steatohepatitis (MASH), an aggressive form of fatty liver disease strongly linked to obesity and Type 2 diabetes. The condition can quietly progress for years before leading to cirrhosis, liver failure or liver cancer. The experimental drug, ION224, blocks a liver enzyme that drives fat buildup and inflammation, two key forces behind liver damage. In clinical trials, patients showed striking improvements in liver health, even without losing weight.

Obesity attacks facial nerves

SCIENTISTS AT Helmholtz Munich and Ludwig Maximilians University Munich (LMU), have created an AI-powered system that can scan and map an entire mouse body in extraordinary detail — and it just uncovered a surprising new effect of obesity. Beyond disrupting metabolism, obesity appears to damage facial sensory nerves linked to touch and sensation, while also triggering widespread inflammation across the body. The study also identified similar molecular patterns in human tissue, suggesting that important aspects of obesity-related nerve damage may occur in both mice and people.

Researchers find a protein that halts brain decline

Low levels of a brain protein called Menin drive ageing & inflammation

SCIENTISTS MAY have uncovered a hidden biological switch that helps control how quickly the body ages. Research published in *PLOS Biology* suggests that declining levels of a brain protein called Menin can trigger inflammation, memory decline and other age-related changes throughout the body. In experiments with mice, restoring the protein reversed several signs of ageing, while a simple amino acid supplement improved cognitive function.

The findings add to growing evidence that ageing may be strongly influenced by the hypothalamus, a small but powerful brain region that regulates metabolism, hormones, body temperature, sleep and stress responses. Researchers increasingly view the hypothalamus as a central command centre for aging itself.

The study, led by Lige Leng and colleagues at Xiamen University in China, focused on Menin, a protein that helps suppress inflammation in the brain. Earlier work had already shown that Menin plays an important role in controlling neuroinflammatory activity. The team wanted to know whether losing this protective protein might contribute to ageing. Their experiments revealed that Menin levels dropped sharply in the hypothalamus as mice grew older. The decline occurred specifically in neurons within the ventromedial hypothalamus (VMH), a region linked to metabolism and systemic ageing. Interestingly, Menin levels did not significantly decrease in nearby support cells such as astrocytes or microglia.

Menin levels dropped sharply in the hypothalamus of lab mice as they grew older. It's a protein that helps suppress inflammation

To investigate what this loss might mean, the researchers engineered mice in which Menin activity could be selectively reduced. The effects were striking. Younger mice with lower Menin levels developed increased brain inflammation, thinning skin, lower bone mass, impaired balance, memory problems and a shorter lifespan compared to normal mice. The results suggest that Menin may act as a protective "anti-ageing" factor inside the brain. One of the most surprising discoveries involved D-serine, an amino acid that also functions as a neurotransmitter in the brain. D-serine helps regulate communication between neurons and is important for learning and memory. When Menin levels fell, D-serine production also dropped. The researchers traced this effect to reduced activity of an enzyme required for D-serine synthesis, which itself appears to be regulated by Menin. D-serine naturally occurs in foods including soybeans, eggs, fish and nuts, and is also sold as a dietary supplement.

The connection caught the attention of researchers because other studies have linked declining D-serine levels with ageing-related cognitive impairment and reduced synaptic plasticity, which means the brain's ability to strengthen neural connections involved in memory and learning.

• CUTTING EDGE

How AI maps are helping India find hidden TB cases before it's too late

Technology is changing how healthcare workers amp up the national TB screening programme

Anonna Dutt

ON THE first floor of a polyclinic in Haryana's Ambala, a health worker logs into the government TB programme's Ni-kshay portal. She clicks on a map of Haryana to check where she should plan her screening camps. All the villages and wards are colour coded red, yellow or green to show the density of the vulnerable population, indicating people who are more likely to have TB. She narrows down on a red patch, a high-risk zone.

This AI-generated map is a new tool in the arsenal of health workers under the National TB Elimination programme. It uses data from nearly 30 open-source geospatial datasets such as population density, nutritional profile, access to health centres — proxies for vulnerability of a population to TB — to create a map of places where health workers are likely to find more TB cases. Called Vulnerability Mapping for Tuberculosis (VMTB), it was piloted in Haryana and Assam before being rolled out across the country under the ongoing 100-day campaign. Smokers, people with diabetes or other respiratory conditions and those who are malnourished are considered to be vulnerable populations for TB. The challenge, however, is that there is no nationwide registry for these conditions. So, how do you find the at-risk population?

"VMTB takes about 30 open-source datasets of socio-demographic and geospatial data. This includes nutrition profiles, NFHS-4 data, presence of night lights and so on. Along with data on historically confirmed TB cases, the AI can predict geospatial vulnerability down to the village and ward level, in fact, down to a 500x500 metre tile," says Dr Neeraj Agrawal, chief programme officer at Wadhwani AI, which developed the programme for the government.

A district programme coordinator uses the tool to identify sites for her next camp

This helps health workers concentrate their resources on areas where more people are likely to get diagnosed during active screening camps, where more cases are likely to be found by testing fewer people. This would ensure early diagnosis of many and fewer people missing from the treatment net. This is essential as the country moves towards eliminating the condition — while India's 2025 deadline has been missed, the global deadline of 2030 remains.

Dr Rajesh Raju, state TB officer for Haryana, says, "There are 6,237 gram panchayats in Haryana. Last year, 35 per cent of these were declared TB-free as they had an incidence of less than one per million population. That leaves us with 4,080 gram panchayats where we would need to focus, something that can be very challenging. This tool has helped us identify 1,968 villages and some more wards, a total of 2,111 high-risk areas where we would be concentrating during the current 100-day campaign. So we can focus 100 per cent of our efforts on 50 per cent of the areas. This has increased our efficiency as well as effectiveness."

This decreased workload has also led to a change in strategy during the current 100-day programme. Instead of testing only the vulnerable populations, the health workers are now screening everyone in these high-risk villages. "With VMTB, now 100 per cent X-ray coverage of the adult population is being attempted in these villages," he adds.

This effort is aided by another AI-technology — a portable hand-held X-ray machine that can screen a person within seconds with an AI-aided tool. Those who show positive indicators for TB have to then give a sputum sample for a confirmatory test. "Our camps look neutral; we do not screen people only for TB. We also conduct routine tests such as BP and blood sugar so that there is less stigma around getting a medical check-up. Getting an X-ray report within minutes — instead of travelling several kilometres to the nearest health centre — is a big draw for patients," says community health officer Avaneeta Sandhu. Patients get their diagnosis the next day. The mapping tool has another use, says Dr Seema Tanwar, civil surgeon, Ambala. "Once the focused activity is over, we can choose the green or yellow villages and screen vulnerable populations to make these villages TB-free."

The tool was trialled in Haryana and Assam. Data from these two states show that there was a 251 per cent relative increase in the yield of TB cases in Haryana and 61 per cent in Assam last year. More than two lakh TB cases were diagnosed, accounting for nearly 10 per cent of the 25 lakh cases detected last year.

Sport

Satwik-Chirag end 2-year title drought with Singapore triumph

Shivani Naik
Mumbai, May 31

Satwiksairaj Rankireddy (L) and Chirag Shetty celebrate after winning the Singapore Open Super 750 on Sunday. BADMINTON PHOTO

IT'S BEEN two long years since Satwiksairaj Rankireddy and Chirag Shetty won a big badminton title. When it finally came in Singapore after a fight back for a 18-21, 21-17, 21-16 victory in 72 minutes, their iconic racquet-twirling and bhangra jigs were back. So was a rocking cradle celebration.

It was also two years of heartbreaking tragedy when Satwik lost his father. On Sunday as they beat World No.3 Fajar Alfian and Muhammad Shohibul Fikri to win a Super 750, Satwik would explain the coordinated baby-celebration.

"Singapore is happy, because after Olympics we have been battling with that (lack of titles). Chasing for that one win. And when we thought we were on the right track, personal things happened. But suddenly one good thing was, last week my brother was blessed with a baby. So the celebration was dedicated to her. We felt like my father was reborn. So she was lucky. This win is dedicated to my family, with brother and mom in the US," Satwik said.

Braving a first set loss and playing the decider for the fourth time this week, the Indians put up a show in fast, furious badminton to win the Super 750, becoming India's first pairing to win Singapore. Their loss at Thailand finals had cast doubts. But their ability to problem-solve and come up trumps even when dragged into a flat game, blazed through this week.

The Indians lost a close first set after 18-18 as the two playmaking Indonesians aced the crisscross exchanges when closing out. But the Indians have hardly been thwarted by these early setbacks this week.

The Indonesians targeted Satwik who was scratchy in defense early on, with even his smashes going into the net. But the Indians have been sticking to their plan of Satwik getting used to front court duties till the halfway point in the match. Fajar-Fikri (FiFa) would also realise that he grows in confidence exponentially as the match progresses.

While the rallies stayed short mostly, the pace of the exchanges was breathless. FiFa are known for their flat crosses and surgical attacks as action remains concentrated on the front court. A 10-second rally could literally have 20 shots in play, and the Indian crouch defense was put to the test.

> ***Braving a first set loss and playing the decider for the fourth time this week, the Indians put up a show in fast, furious badminton.***

Once Chirag stepped forward, the Indians picked up pace, but Satwik was also finding bissecting cross drives, even as the Indian serves swung between faults and fabulous.

"These matches you don't get so many smashes to hit," Satwik told BWF. "So you have to prove yourself you can play at the net as well. My job is to serve and catch first shot then hand it over to Chirag. I felt they were under pressure when we were chasing up and playing very much simple. When a few net chords helped us I thought today is our day. I was so happy."

Opening up a 18-15 lead in the second set gave Indians the impetus and deflated Fikri who made a bunch of rush-of-blood errors. FiFa were anyway struggling to keep shuttle in bounds on the backlines. But coach Tan had also told Satwik-Chirag to be alert to the first three shots.

"With Indonesians, Malaysians it's mainly about first four shots," Satwik told BWF. "If they keep us under pressure they are on top, they are very good in those first 4 shots. So coach said don't worry about back court, focus on front court."

In the finishing stages of the second, Satwik found the line on a lofted shot and pushed a decider with a lucky net chord. The Indian attack by then had composed itself — absorbing pressure of the flat crisscrosses in defense, and pouncing on lifts given Satwik smashes were impossible to defend for the Indonesians.

Even as the Indonesians correctly attacked Satwik chasing him like a tracker about the court, he came up with the best point of the Singapore smashing success. On a dipping shuttle in front of him, when it was inches off the ground, he curled his wrists exquisitely with the racquet head pointing down to send the swishing lift to the back line for 9-5 in the decider.

When Fikri got going and reduced gap to 13-15, the Indian soaring attack was back. Three smashes from Chirag got dug out by Alfian, but they had no answer to the 4th from Satwik.

At 16-13 and 20-15, Satwik unleashed a hat-tip to India's obsessed sport — cricket. With his version of yorker-smashes. No matter how good your defense — and the Indonesians have very good retrieving capabilities — there's little you can do when the smash is aimed right at the toes. Alfian could only sweep it away on the floor — such was the precision. Satwik snuck in a yellow card in the middle of all this for not being ready quick enough to receive serve. But his last down hit was sent flying wide by a grimacing, glum Alfian.

"I've put in a lot of effort and energy in my celebration. Jokes aside, we knew the wins would come. Just a matter of time," Chirag said.